

MARITIME SAFETY COMMITTEE
109th session
Agenda item 5

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**DEVELOPMENT OF A GOAL-BASED INSTRUMENT
FOR MARITIME AUTONOMOUS SURFACE SHIPS (MASS)**

Report of the Intersessional MASS Working Group

SUMMARY

<i>Executive summary:</i>	This document provides the report of the third session of the Intersessional MASS Working Group.
<i>Strategic direction, if applicable:</i>	2
<i>Output:</i>	2.23
<i>Action to be taken:</i>	Paragraph 34
<i>Related document:</i>	MSC/ISWG/MASS 3/WP.1

INTRODUCTION

1 The Intersessional Working Group on Maritime Autonomous Surface Ships (the Group) (MSC/ISWG/MASS), chaired by Mr. Henrik Tunfors (Sweden), held its third session from 9 to 13 September 2024.

2 The Group was attended by delegations from the following Member States:

ARGENTINA	INDIA
AUSTRALIA	INDONESIA
BAHAMAS	ITALY
BELGIUM	JAPAN
BRAZIL	KENYA
CANADA	LIBERIA
CHILE	MALAWI
CHINA	MALAYSIA
CYPRUS	MARSHALL ISLANDS
DENMARK	MEXICO
EGYPT	NETHERLANDS (KINGDOM
EL SALVADOR	OF THE)
FINLAND	NIGERIA
FRANCE	NORWAY
GERMANY	PAKISTAN

PANAMA
PARAGUAY
PERU
POLAND
QATAR
REPUBLIC OF KOREA
RUSSIAN FEDERATION
SAUDI ARABIA

SINGAPORE
SOUTH AFRICA
SPAIN
SWEDEN
TÜRKİYE
UNITED ARAB EMIRATES
UNITED KINGDOM
UNITED STATES

by a representative from the following Associate Member of IMO:

HONG KONG, CHINA

by observers from the following intergovernmental organizations in consultative status:

EUROPEAN COMMISSION (EC)
MARITIME ORGANISATION OF WEST AND CENTRAL AFRICA (MOWCA)
INDIAN OCEAN MEMORANDUM OF UNDERSTANDING ON PORT STATE
CONTROL (IOMoU)

and observers from the following non-governmental organizations in consultative status:

INTERNATIONAL CHAMBER OF SHIPPING (ICS)
BIMCO
INTERNATIONAL ASSOCIATION OF CLASSIFICATION SOCIETIES (IACS)
OIL COMPANIES INTERNATIONAL MARINE FORUM (OCIMF)
INTERNATIONAL MARITIME PILOTS' ASSOCIATION (IMPA)
INTERNATIONAL FEDERATION OF SHIPMASTERS' ASSOCIATIONS (IFSMA)
CESA
INTERNATIONAL ASSOCIATION OF INDEPENDENT TANKER OWNERS
(INTERTANKO)
INTERNATIONAL ASSOCIATION OF DRY CARGO SHIPOWNERS
(INTERCARGO)
EUROPEAN ASSOCIATION OF INTERNAL COMBUSTION ENGINE AND
ALTERNATIVE POWERTRAIN MANUFACTURERS (EUROMOT)
THE INSTITUTE OF MARINE ENGINEERING, SCIENCE AND TECHNOLOGY
(IMarEST)
INTERNATIONAL MARINE CONTRACTORS ASSOCIATION (IMCA)
INTERNATIONAL TRANSPORT WORKERS' FEDERATION (ITF)
WORLD SHIPPING COUNCIL (WSC)
THE NAUTICAL INSTITUTE (NI)
ACTIVE SHIPBUILDING EXPERTS' FEDERATION (ASEF)
ZERO EMISSIONS SHIP TECHNOLOGY ASSOCIATION (ZESTAs)
ONE SEA RY

TERMS OF REFERENCE

3 The Group was instructed, taking into account the progress made by the comments and decisions made at MSC 108, to:

- .1 finalize the development of part 3, based on annex 1 to document MSC 108/WP.7;

-
- .2 if time permits, finalize the following chapters in part 2:
- .1 Approval process;
 - .2 Risk assessment;
 - .3 System design;
 - .4 Software principles;
 - .5 Management of safe operations;
 - .6 Connectivity;
 - .7 Radiocommunications; and
 - .8 Maintenance and repair;
- .3 invite GBS expert(s) to provide expert advice for the work under sub-paragraphs .1 and .2; and
- .4 submit a written report to MSC 109.

OUTCOME OF MSC 108 AND MASS-JWG 3

4 The Group observed with appreciation a presentation delivered by the Secretariat outlining the outcome of MSC 108 and MASS-JWG 3, summarizing the information provided in document MSC/ISWG/MASS 3/2 (Secretariat), which can be accessed on IMODOCS (MSC/ISWG/MASS 3/Virtual Portal) and downloaded [here](#).

REVIEW OF THE CODE BY THE GBS EXPERT(S)

5 Regarding the matter of inviting GBS expert(s) to provide a review and advice on the draft MASS Code, as instructed by MSC 108, the Group noted the information presented by the Secretariat indicating that the task was ongoing with the invited GBS expert(s) and a work plan would be confirmed after MSC/ISWG/MASS 3, with a view to providing input to MSC 110. The GBS expert(s) review is expected to ensure consistency across different chapters of the draft MASS Code, taking into account the result of this session of the Group, as well as the ongoing work of the MASS Correspondence Group.

STATUS OF THE MASS CORRESPONDENCE GROUP

6 The Group noted with appreciation the oral report provided by the Coordinator* of the MASS Correspondence Group with regard to the status of the development of the draft MASS Code; and noted that the Group had made further progress and that it would submit its report to MSC 110, taking into account the outcome of this session of the Intersessional MASS Working Group, as well as the relevant outcome of MSC 109.

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FURTHER DEVELOPMENT OF THE DRAFT MASS CODE

7 The Group agreed to use document MSC/ISWG/MASS 3/WP.2 as a basis for its consideration and further development of the draft MASS Code.

Consideration of part 3 of the draft MASS Code***Chapter 17 (Safety of Navigation)***

8 The Group considered, further developed and finalized chapter 17 (Safety of Navigation) of the draft MASS Code, taking into account the proposals in documents MSC/ISWG/MASS 3/3/1 (IHO), MSC/ISWG/MASS 3/3/2 (Japan et al.) and MSC/ISWG/MASS 3/3/3 (Australia et al.), as set out in the annex.

General

9 With regard to section 17.3.1, and in addition to the Expected Performance (EP) providing that non-navigation functions or tasks should not endanger the safe operation of an Autonomous Navigation System (ANS) or system for remote navigation, the Group agreed to an additional EP in relation to effective integration and coordination with other ship systems to avoid conflicting operational commands.

10 With respect to data format and availability, the Group considered the specific proposals in documents MSC/ISWG/MASS 3/3/1 and MSC/ISWG/MASS 3/3/3; and:

- .1 noted that data availability, format and usability matters should be applicable to the whole chapter, and subsequently agreed to new Functional Requirements (FRs) and associated EPs accordingly (17.3.2);
- .2 agreed that the important aspect was to ensure the availability of necessary data for an ANS or system for remote navigation in a machine-readable format, but that it should not specify any reference to a specific standard for the data necessary for safe navigation;
- .3 agreed not to include an EP that an ANS or system for remote navigation should be provided in a human-readable format in this chapter, as this should be covered in section 9.2 (Human Centred Design); and
- .4 agreed to insert an EP on the use of electronic means for some information, addressing a potential gap for a MASS below 3,000 gross tonnage with respect to the carriage requirement of charts and publications (SOLAS regulations V/19.2.1.4 and 19.2.1.5).

11 Further, with regard to the issue of proper records relating to navigational data for the safety of navigation and casualty investigation purposes, the Group:

- .1 acknowledged the relevance of such a functionality also for MASS, not only for navigation but also for other functions; and
- .2 agreed that such provisions on record keeping could be included in part 2 of the draft MASS Code, and, therefore, invited the Committee to instruct the MASS Correspondence Group to address this, as appropriate.

Sub-functions for MASS navigation

12 The Group agreed that the FRs and EPs required consistency across this section, and thus adopted a structured approach by first identifying the respective sub-functions needed for a MASS navigation system, then formulating the relevant FRs and subsequently the associated EPs. The Group found this approach particularly useful and, therefore, consideration should be given to adopting this for future reviews of the Code, as appropriate.

13 The Group subsequently considered and agreed on the following sub-functions for MASS navigation: voyage planning, situational awareness, collision and grounding avoidance, and route execution and monitoring. Accordingly, the Group prepared FRs and EPs for these sub-functions, as outlined below.

Voyage planning

14 The Group considered whether reference should be made to SOLAS regulation V/34 regarding voyage planning; but agreed to refer to resolution A.893(21) on *Guidelines for voyage planning* only, as it added additional guidance, while SOLAS was already referenced in the FRs of chapter 17.

15 Additionally, the Group considered the EPs for the cases when the voyage plan was prepared by the ANS/system for remote navigation itself or by an external source; and:

- .1 agreed that any ANS or system for remote navigation should be capable of receiving the voyage plan in all cases. However, generating such plans on its own should be optional, taking into account various systems available on the market. Therefore, the Group established additional EPs for those systems generating the voyage plan themselves and drafted two separate sets of FRs and EPs accordingly;
- .2 recalled that the master had to approve the voyage plan before use;
- .3 agreed that the ANS/system for remote navigation should ensure full control and access of the master for approval, review or modification of the voyage plan; and
- .4 agreed that the ANS/system for remote navigation should include relevant functionalities giving confirmation of correct reception and usability of the voyage plan.

Situational awareness

16 The Group discussed the concepts associated with situational awareness and agreed on an FR and relevant EPs addressing monitoring, integration and analysis, and projection or prediction of future statuses.

17 In the context of the human-machine interface and accessibility to situational awareness results, the Group considered how to indicate the role and responsibilities of the master, officer of the watch (OOW) or "responsible person". The Group acknowledged the unique, overarching role and responsibility of the master and that the master could delegate some of the tasks and duties when necessary. In this regard, the Group agreed to make references to the master only; however, it acknowledged that there might be a need for separate provisions on delegation of tasks and duties elsewhere in the draft MASS Code to address this potential gap. Therefore, the Group agreed to invite the Committee to consider this.

Collision and grounding avoidance

18 With respect to collision and grounding avoidance, the Group agreed that an ANS or system for remote navigation should take action in accordance with the Convention on the International Regulations for Preventing Collisions at Sea, 1972 (COLREG) to prevent collisions, but that the MASS Code should also address allisions and groundings, without causing contradiction with existing requirements in COLREG. Subsequently, the Group agreed to an FR and EPs for collision and grounding avoidance.

Route execution and monitoring

19 When considering the sub-function for route execution and monitoring, the Group:

- .1 agreed that the execution of a planned route was encompassed by the "voyage plan" as the overarching term, and that the term "passage plan" was a part of the voyage plan; and
- .2 agreed not to make reference to restrictions (e.g. areas to be avoided) and local regulations for route execution purposes, noting that such concepts were already captured in the section for "Voyage planning".

The Group subsequently agreed to an FR and EPs for route execution and monitoring.

Override and fallback response

20 The Group, when considering override and fallback response, including the relevant proposals in document MSC/ISWG/MASS 3/3/2;

- .1 agreed to remove references to fallback state and retained only the "override" function, as the fallback state was already captured in chapter 8 (Operational context) of the draft MASS Code; and
- .2 acknowledged that "override" might also be relevant for functions other than navigation; and agreed that the term could be defined in section 4 (Terminology and definitions). Therefore, the Group invited the Committee to instruct the MASS Correspondence Group to address this, as appropriate.

Chapter 20 (Fire Protection, fire detection and fire extinction)

21 The Group considered, further developed and finalized chapter 20 (Fire Protection, fire detection and fire extinction) of the draft MASS Code, taking into account the proposals in document MSC/ISWG/MASS 3/3 (Norway), as set out in the annex.

22 When further developing the chapter, the Group:

- .1 agreed that a fire in a Remote Operations Centre (ROC) should be addressed in chapter 18 (Remote operations) insofar as it related to the effect on the remote operation of the MASS and that the physical facility of the ROC would also be subject to local regulations in fire response and preparedness;

- .2 considered the requirement for a ship to remain under control during a fire event and agreed that a fire outside of a compartment that was directly linked to the control of the ship should not result in a loss of control, however that a fire within such a compartment might require the ship to enter a fallback state. Subsequently, the EPs were revised to clarify this distinction;
- .3 agreed on an EP under 20.2.3 requiring fire detection in all spaces on board with a fire risk, including e.g. compartments and open spaces, to be provided with a means to detect the fire;
- .4 included an EP to confirm a fire using a different detection source (20.2.3);
- .5 agreed that information and instructions in relation to fire safety should be provided to any personnel boarding the ship (20.2.4);
- .6 addressed the safe isolation of fire-extinguishing systems for compartment access and provision of onboard indication prior to, and during, activation (20.2.4);
- .7 agreed that the stability of the ship should be monitored during any fire-fighting operation not only with respect to onboard water-based fire systems but also the external fire-fighting response (20.2.4);
- .8 agreed that fire growth should be monitored and assessed (20.2.5); and
- .9 made a reference to "containment measures" in the FR on active fire protection measures, given the importance of containment in fire-fighting (20.2.6).

Chapter 23 (Search and rescue)

23 The Group considered chapter 23 (Search and rescue) of the draft MASS Code, taking into account the proposals included in document MSC/ISWG/MASS 3/3.

24 In particular, the Group considered the implications for MASS with regard to rendering assistance to ships and persons in distress at sea. During the discussion, the Group noted the following views:

- .1 a pragmatic approach should be taken to prevent the relaxation of the obligations without imposing safety measures beyond what would be expected for conventional ships;
- .2 a MASS should not be exempt from the requirements of SOLAS regulation V/33;
- .3 a MASS having advanced technology, e.g. sensors for detecting survivors at sea, could perform or contribute to search and rescue operations;
- .4 search and rescue had different implications for MASS of different sizes and different operational concepts, and the matter should, therefore, be addressed through high-level goals and functional requirements; and
- .5 in relation to SAR, there are obligations under various IMO instruments, such as the SAR Convention, as well as other international instruments under the remit of other Organizations, e.g. the IAMSAR Manual and UNCLOS.

25 Consequently, noting the need for further discussion on the matter, the Group invited the Committee to further consider this chapter at the next session, taking into account document MSC/ISWG/MASS 3/3.

Chapter 28 (Emergency response)

26 The Group considered chapter 28 (Emergency response), taking into account document MSC/ISWG/MASS 3/3, proposing the deletion of the entire chapter, given that the operational elements of the chapter were sufficiently covered in existing instruments (e.g. the ISM Code, Part A, 8 and resolution A.852(20)) and other chapters of the draft MASS Code.

27 Having considered the arguments presented in document MSC/ISWG/MASS 3/3, and noted that additional submissions could be made to MSC 109, the Group agreed that this matter should be considered at MSC 109.

28 Subsequently, the Group invited the Committee to further consider chapter 28 at the next session, taking into account document MSC/ISWG/MASS 3/3 and the discussion at this session.

DISSEMINATION OF THE HAZID TABLES

29 The Group noted that the hazard identification (HAZID) tables developed by the splinter group leaders would be essential to understand the drafting of the FRs and EPs, but recognized that these might not reflect changes made to the draft MASS Code since MSC 108.

30 In this respect, the Group invited the leaders to share the latest available HAZID tables with the IMO Secretariat (mass@imo.org), with a view to them being made available on IMODOCS (MSC/ISWG/MASS 3/Virtual Portal) for future reference.

REMAINING MATTERS

31 Due to time constraints, the Group was not able to consider chapters 6, 7, 9, 10, 11, 12, 13 and 16 in part 2 and chapters 18, 19, 21, 22, 24, 25, 26 and 27 in part 3 of the draft MASS Code, as well as the relevant proposals in documents MSC/ISWG/MASS 3/3 and MSC/ISWG/MASS 3/4 (China).

32 In this respect, the Group invited the Committee to consider at the next session the above-mentioned documents and those proposals that have not been addressed by the Group, possibly through a working group, if established. Therefore, the Group prepared the following draft terms of reference related to the outcome of this session, for consideration:

"The MASS Working Group, taking into account the discussions of MSC/ISWG/MASS 3, is invited to consider chapters 6, 7, 9, 10, 11, 12, 13 and 16 in part 2 of the draft MASS Code, as well as chapters 18, 19, 21, 22, 23, 24, 25, 26, 27 and 28 in part 3 of the Code, together with the relevant proposals in documents MSC/ISWG/MASS 3/3 and MSC/ISWG/MASS 3/4".

33 One delegation expressed the view that based on the progress made during this meeting, it would be highly unlikely that the draft MASS Code would be finalized at MSC 110 as currently scheduled. In the discussion that followed, it was agreed to further consider the timeline towards finalization and also the possible need for intersessional meetings in 2025 during the deliberations of the MASS Working Group, if established.

ACTION REQUESTED OF THE COMMITTEE

- 34 The Committee is invited to approve the report in general, and in particular to:
- .1 note that the Group finalized chapters 17 and 20 of the draft MASS Code, and had initial discussions on chapters 23 and 28 (paragraphs 7 to 28 and annex);
 - .2 instruct the MASS Correspondence Group to address the potential contributions identified by the Group, as appropriate (paragraphs 11.2 and 20);
 - .3 consider addressing a potential gap in the draft MASS Code on delegation of the master's tasks and duties in the context of MASS (paragraph 17);
 - .4 invite the leaders of the splinter groups that contributed to the development of the draft MASS Code to share the latest available HAZID tables with the IMO Secretariat, with a view to them being made available on IMODOCS (MSC/ISWG/MASS 3/Virtual Portal) for future reference (paragraph 30);
 - .5 note that due to time constraints, the Group could not (paragraphs 25, 28 and 31):
 - .1 consider chapters 6, 7, 9, 10, 11, 12, 13 and 16 in part 2, and chapters 18, 19, 21, 22, 24, 25, 26 and 27 in part 3 of the draft MASS Code, as well as the relevant proposals made in some of the documents submitted to this session; and
 - .2 finalize the discussion on chapters 23 and 28; and
 - .6 establish a working group to further progress the work, taking into account the draft terms of reference prepared by the Group (paragraph 32).

ANNEX*

NOTE: The proposals in the submissions made to MSC/ISWG/MASS 3 that have not been discussed, are kept in this version for ease of future reference and discussion.

CONSOLIDATED VERSION OF THE DRAFT INTERNATIONAL CODE OF SAFETY FOR MARITIME AUTONOMOUS SURFACE SHIPS (MASS CODE)

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* This annex is provided in the English language only. Colour shading has been used for indicating particular reminders and additional modification proposals to the draft text, emanating from the submissions made to this session and the proposals by the Secretariat.

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PREAMBLE

1. Existing IMO instruments have historically been developed on the basis that the ship will have at least a minimum level of manning on board to carry out the various tasks required to ensure safe, secure, and environmentally sound ship operations.
2. The ever-increasing use of automation in the operation of ships, along with the anticipated increase in the use of remote control and autonomous operation of key functions, will require a different approach and therefore some adjustment of the accepted norms regarding on board manual intervention and control as [contained] [reflected] within SOLAS and other IMO instruments.
3. In facing these challenges it is recognized that some aspects associated with MASS are not adequately or fully addressed in SOLAS or other IMO instruments and that additional guidance is required on the design and operation of MASS to ensure a level of safety that is ~~equivalent to that~~ expected of a conventionally operated ship.
4. [The Code and the use of MASS are required to conform to the relevant rules of international law, including the United Nations Convention on the Law of the Sea (UNCLOS), and generally accepted international regulations, procedures and practices developed by the International Maritime Organization (IMO) as the competent international organization for global shipping.] [UAE proposal in LEG 111/10/5, par.8]
5. This Code addresses the functions needed for safe, secure, and environmentally sound operations of MASS insofar as they are not adequately or fully addressed in other applied IMO instruments, while ensuring that required safety levels are maintained when implementing remote controlled or autonomous operation of key functions.
6. This Code is intended as supplementary to other IMO instruments, such as SOLAS, and provides a regulatory framework for remotely controlled and autonomous operation of key functions.
7. The safety principles and objectives of this Code reflect changes in the operational risks [(increases or reductions)] which may result from the introduction of remote control and autonomous operation of key functions and address their management and reduction through mitigation measures and controls.
8. This Code has been developed based on the *Generic guidelines for developing IMO Goal-based Standards* (MSC.1/Circ.1394/Rev.2) and the *Principles to be considered when drafting IMO instruments* (resolution A.1103(29)).
9. [The provisions of this Code should be implemented for individual remotely controlled or autonomous functions even where persons are on board to handle other functions.]
10. [This Code takes into account that certain operational functions may be controlled from a location, or locations, remote from the MASS and addresses necessary aspects of such Remote Operations Centres (ROCs).]

- [10*bis* Advancing technology in the shipping industry leads to an ever-increasing use of automation to operate ships. More automation does not qualify a ship as a Maritime Autonomous Surface Ship (MASS). The main qualifier to distinguish a MASS from a conventional ship is that some or all functions of the ship, as performed by humans, both aboard and ashore, are augmented by advanced automation and remote operations.]
11. [Enhanced automation does not qualify a ship as a MASS. The main qualifier to distinguish a MASS from a conventional ship is the introduction of autonomous or remote operation technology augmenting or replacing functions performed by seafarers on board involved in conducting or controlling these ship functions.]

PART 1 INTRODUCTION

CHAPTER 1 PURPOSE, PRINCIPLES AND OBJECTIVES

1.1 Purpose

The purpose of this Code is to provide an international regulatory framework to address both, remote control and autonomous operation of key functions and ensure safe, secure, and environmentally sound operation of Maritime Autonomous Surface Ships (MASS). The Code further aims to support the safe and secure adoption and integration of new technology for ship operations and provide for consistency of approach to the design, build and operation of MASS.

1.2 Principles

This Code is developed on the principles that it be:

- a) supplementary to any applied base instruments, such as SOLAS, and only address MASS functions as far as they are not adequately or fully addressed in the applied base instruments;
- b) holistic to ensure the objectives, aims and principles of the base instruments are maintained whilst also enabling the MASS functions and operations to be addressed across all instruments;
- c) goal-based and addressing matters at the functional level;
- d) non-mandatory although developed in such a way as to facilitate future transition to mandatory status; and
- e) technology neutral acknowledging industry practices and experience in the deployment of new technologies.

[N.B. *Location to be confirmed:*

The Code is based on the following principles:

- [.1 there should be a human master responsible for a MASS, regardless of mode of operation;
- .2 a master of a MASS may not need to be on board, depending on the technology used on the MASS and human presence on board, if any; and
- .3 regardless of mode of operation, the master of a MASS should have the means to intervene when necessary.
- .4 several masters may be responsible for a MASS on a single voyage, under certain conditions, and that only one master should be responsible at any given time (further consideration of what those conditions are is required).]]

1.3 Objectives

In achieving its Purpose, this Code is intended to:

- a) prevent relaxation of the level of accepted standards for design, construction, or operation and ensure a level of safety of that expected of a conventional ship;
- b) enable all ships to safely coexist without impeding or negatively impacting each other, regardless of whether certain functions are remotely controlled or autonomously operated;
- c) allow for the application of solutions that are demonstrably safe, secure, and environmentally sound in performing the designated functions in all defined conditions; and
- d) be cognizant of the potential for the unintended placement of regulatory barriers to new or novel application of remote control or autonomous technology on ships.
- e) provide for human oversight and control of MASS operation and MASS functions; and
- f) address responsibility for MASS.

CHAPTER 2 APPLICATION

This Code applies to cargo ships to which SOLAS chapter I applies which have functions that enable autonomous or remote operations including any associated ROC(s) when the Administration deems that compliance with base instruments is not practicable.

This Code does not apply to:

- .1 cargo high speed craft to which SOLAS chapter X applies; and
- .2 warships, naval auxiliaries, and other ships owned or operated by a Contracting Government and used only on Government non-commercial service.

CHAPTER 3 CODE STRUCTURE

The structure and intent of the Parts of this Code are:

- Part 1: Introduction covering overarching matters to be considered in the application of the Code.
- Part 2: Technical principles applicable in all cases when applying this Code to autonomous or remotely operated functions. These principles and requirements should be met as part of any approval and certification process.
- Part 3: Goals, functional requirements, and provisions applicable to autonomous or remotely operated functions. Depending on the mode of operation and functionality being certified, not all chapters of Part 3 may require to be met.
 - [The term 'ship', unless expressly provided otherwise, refers to a ship to which the MASS Code applies.
 - Whenever the term "ship" is used in this Code, it should be read as ship to which the MASS Code applies.]

CHAPTER 4 TERMINOLOGY AND DEFINITIONS

For the purposes of the Code, unless expressly provided otherwise, terms used have the meanings defined in the following paragraphs. For terms used, but not defined in this Code, the definitions as given in the 1974 SOLAS Convention, as amended, shall apply.

Note - Following feedback from multiple delegations, it is felt that certain terms are included which are not required as they are either copied or defined already in other IMO instruments and do not add value. Therefore, any terms in italics indicate those which are defined elsewhere may be removed at a later stage.

4.1 Administration

Administration means the Government of the State whose flag the MASS is entitled to fly.

[4.1bis Alarm – to be defined.]

[4.1ter Alert – to be defined.]

4.2 [[Annunciated] [Announced] failure

An *annunciated failure* is one which fails "actively", i.e., in such a manner as to inform crew of the failure by virtue of system generated cues such as visual and/or audible notifications, warnings, and alarms. (RBAT)

or

[An *annunciated failure* is the situation when a failure is accompanied by information to the crew of the failure by virtue of system generated cues such as visual and/or audible notifications, warnings, and alarms]]

4.3 [Approved

Approved means approved by the Administration.]

4.4 [Automated] [Autonomous] functions

[*Automated functions* means automated processes, parts of the system that may be automated when it is not the ship being considered as one whole. [Note - Automated systems was proposed instead of functions but consensus was on functions at this time.]

or

Autonomous functions are functions (or complete ships) that may operate in complex and open-ended environments with high levels of independence and self-determination. They perceive, learn, reason and [act with self-awareness and] respond [intelligently] [appropriately] to unforeseen changes in the environment. (Denmark proposal from 1.2 (Application)).]

4.5 [Automatic

Automatic means processes or equipment that, under specified conditions, can function without human control. (RBAT)]

4.6 Autonomous

Autonomous means processes or equipment in a MASS system which, under certain conditions, are designed and verified to be controlled by automation, without human assistance. (RBAT)

4.7 Autonomous Navigation System

[Autonomous Navigation System (ANS) means a system which has the functionalities of situational awareness, route planning and determination for collision and grounding risk avoidance, ship's heading, speed and track control, etc. (MSC 107/5/10)

or

Autonomous Navigation System (ANS) means a set of elements that provide functions related to autonomous navigation within a defined or higher operational envelope. It also should include the possibility of remote control. (MSC 107/5/7)]

4.8 *[Cargo Ship*

Cargo Ship means any [full or semi-displacement] ship which is not a passenger ship, a ship of war and troopship, [or] a ship which is not propelled by mechanical means, [or] a wooden ship of primitive build, [or] a fishing vessel or a mobile offshore drilling unit. (2008 IS Code)]

[4.8bis COLREG

COLREG means the Convention on the International Regulations for Preventing Collisions at Sea (COLREG), 1972.]

4.9 *[Company*

Company means the owner of the MASS ship or any other organization or person such as the manager, or the bareboat charterer, who has assumed the responsibility for operation of the ship from the shipowner and who, on assuming such responsibility, has agreed to take over all the duties and responsibilities imposed on the Company by the MASS Code.]

4.10 Concept of Operation (ConOps)

ConOps means a document describing the characteristics of a proposed system. The ConOps would be part of the certification of a as MASS.

4.11 *[Control function*

Control function means actions performed by humans or software for the accomplishment of a functional goal (adapted from IEC, 2000).]

4.12 *[Control action*

Control action means the acquisition of information, analysis of information, decision-making, or implementation of physical actions performed as part of a control function.]

4.12bis. Degraded state

Degraded state means a [divergence/deviation] in the normal operation which can potentially result in a fallback state.

Proposal by IHO (MSC/ISWG/MASS 3/3/1) – NOTE: Proposal not discussed at MSC/ISWG/MASS 3 – included for ease of future reference:

4.12ter. Electronic Navigational Data Service

Electronic Navigational Data Service means all information, necessary for safe and efficient navigation, originated by, and distributed on the authority of, government authorized hydrographic service as required by regulations V/19 and V/27 of the SOLAS Convention. ENDS encompasses nautical chart and nautical publication data as defined in SOLAS Chapter V and IHO standards in force.

4.13 Failure

Failure means the termination of the intended behaviour of an element or item due to fault manifestations. (MSC 107/5/7)

or

Failure means the loss of the ability of an item to perform the required (specified) function within the limits set for its intended use. (RBAT)

4.13.bis Fallback response

Fallback response means the actions and procedures to enter into, safely stay within, and eventually recover from, a predefined fallback state.

4.14 Fallback state

Fallback state means a designed state that can be entered through a fallback response when it is not possible for the MASS with its autonomous or remote-controlled ship functions to stay within the operational envelope.

4.15 Fault

Fault means an abnormal condition that [may cause a reduction in, or loss of, the capability of a functional unit to perform the required function] or [can cause an element or an item to fail]. (MSC 107/5/7)

4.16 Function

Function means a group of tasks, duties and responsibilities, as specified in the MASS Code, necessary for MASS operation, safety of life at sea, [security of the vessel] or protection of the marine environment.

4.17 Functional analysis

Functional analysis means the examination of the functional goals of a system with respect to available manpower, technology, and other resources, to provide the basis for determining how the function may be assigned and executed (IEC, 2009).

4.18 [High-Speed Craft

High-speed craft (HSC) means a craft capable of a maximum speed, in metres per second (m/s), equal to or exceeding: $3.7 \cdot \nabla^{0.1667}$ where: ∇ = displacement corresponding to the design waterline (m³). (2008 IS Code)]

4.19 Human-Automation interaction

Human-Automation interaction means the way a human [performs a control function or] is affected by, controls, and receives information from automation while performing a task (Sheridan & Parasuraman, 2006)

4.20 [Human Element

Human Element means the interaction between the autonomous systems and the human operators involved in the operation and management of MASS. [These factors should, amongst others, include cognitive workload, situational awareness, communication protocols, teamwork, decision-making processes, training requirements for human operators as well as guidelines and best practices to ensure that these factors are adequately addressed in the design and operation of MASS.]]

[4.20bis Human Machine Interface (HMI) [to be defined].]

[4.20ter Human Centered Design *[to be defined]*]

4.21 In service (operating, under remote operation, under remote supervision; need to cover in dry dock [ready to operate]) - to be defined.

4.22 *[International Convention on Maritime Search and Rescue*

International Convention on Maritime Search and Rescue (SAR), 1979, as amended.]

4.22 *[International Safety Management (ISM) Code*

International Safety Management (ISM) Code means the International Management Code for the Safe Operation of Ships and for Pollution Prevention as adopted by the Assembly, as may be amended by the Organization.]

4.23 [Intolerable risk

[Intolerable risk means the level of risk at individual and societal level that would be higher than ALARP (As Low As Reasonably Practicable) assessed during the design [of ConOps for] [Mission] of MASS.] (India proposed from MSC 91-16).]

4.24 *[MARPOL*

MARPOL means the International Convention for the Prevention of Pollution from Ships, 1973, as modified by the Protocol of 1978 relating thereto as amended by the 1997 Protocol.]

4.25 Maritime Autonomous Surface Ship (MASS)

Maritime Autonomous Surface Ship (MASS) means a ship which, to a varying degree, can operate independent of human interaction.

4.26 [MASS Onboard Crew

MASS onboard crew means a master, other officers and operational staff [physically][who may be present] on board a MASS.]

4.26bis [MASS operations

MASS operations means operations of MASS on board or by ROC.]

4.27 [MASS Remote Crew

MASS remote crew means a remote master, remote operators and responsible persons controlling operating MASS remotely and/or providing assistance to the crew in the MASS operation.]

4.28 Master/master of a MASS

Master [of a MASS] means the person having command of a MASS ship (STCW)

[4.28bis Minimal Risk Manoeuvre (MRM) - to be defined if used in Code]

4.29 Mission

Mission means the commercial, political or public intentions which have contributed to and justifies the vessel concept development and operation.

4.30 [Mission phase

Mission Phase means the subdivisions of the mission typically characterized by a recognizable shift in where the vessel is located in terms of geographical surroundings, or the start and end of one or more operations.]

4.31 Mitigation

Mitigation means a measure implemented to prevent unsafe conditions or modes from resulting in losses.

4.32 Mitigation layer

Mitigation layer means a mitigation capable of preventing a scenario from proceeding to an accident without being adversely affected by the initiating event or the action of any other mitigation layer associated with the scenario.

4.33 Modes of Operation

Modes of Operation means the condition(s) under which the functions of a MASS are controlled, i.e. remote-control or autonomous with or without persons on board.

[4.33bis. Normal operation

Normal operation means ship operations within the Operational Envelope of a MASS, where the ship is able to continue sailing even at a degraded state.]

[4.33ter [Operational Design Domain (ODD)]

[*Operational Design Domain (ODD)*] means a document providing the conditions, related control modes and modes of operation under which any individual autonomous or remote-operated ship function is designed to operate, including all tolerable events]

[4.33ter Operator failure]

Operator Failure means a situation where an operator either fails to take an appropriate action or takes incorrect actions in a particular context that could be driven by numerous factors including limited or poor-quality information, insufficient time to respond effectively, and a lack of understanding regarding the proper course of action to take.]

4.34 Operational Envelope

The *Operational Envelope* should provide ship's operational capabilities and limitations and ship-specific capabilities and limitations.

4.35 [Organization

Organization means the International Maritime Organization.]

[4.35bis Quality of Service - to be defined if used in the Code]

4.36 Process

Process means a set of interrelated or interacting activities that transforms inputs into outputs (IEC, 2018)

4.37 Remote Control:

Remote control is when the ship, or functions within the ship, are operated from outside the [controller area network of the] ship without interference from anyone on board the ship. Remote control may have direct control of actuators on board, or may just give functional commands to an autonomous function (system). Remote control may have varied complexity, from simple communication of setpoints to full real time control including full virtual feedback from the ship/function. (Denmark suggestion from 1.2 (application))

- Remote Control Station means a system connected to MASS for its remote control. (MASS-JWG1/WP.1)

- Control stations are those spaces in which the ship's radio or main navigating equipment or the emergency source of power is located or where the fire recording or fire control equipment is centralized.' (SOLAS chapter II – 18)
- Control and monitoring equipment means the equipment installed for the effective operation and control of the BWMS and the assessment of its effective operation. (Ballast Water Management System (BWMS) Code)
- Control Station are those spaces in which the craft's radio or navigating equipment (main displays and controls for equipment specified in 13.2 to 13.7) or the emergency source of power and emergency switchboard are located, or where the fire recording or fire control equipment is centralized, or where other functions essential to the safe operation of the MASS craft such as propulsion control, public address, stabilization systems, etc., are located. (High Speed Craft Code)
- Operating station means a confined area of the operating compartment equipped with necessary means for navigation, manoeuvring and communication, and from where the functions of navigating, manoeuvring, communication, commanding, conning and lookout are carried out.' (High Speed Craft Code)
- Control station means a single or multiple position including all equipment such as computers and communication terminals and furniture at which control and monitoring functions are conducted. (ISO 11064-3)
- Remote Control Station means a place from which MASS, or functions of a MASS can be operated. A ROC may have multiple control stations within its facilities.' (MASS Code Remote Operation Section 3.2)

4.38 Remote Operator

Remote Operator means a qualified person who is employed or engaged to operate some or all aspects of the functions of a MASS from a Remote Operations Centre.

4.39 Remote Operations Centre (ROC)

Remote Operations Centre (ROC) means a location remote from the MASS that can operate some or all aspects of the functions of the MASS.

4.40 Remote Master

Remote Master means a master who is in a Remote Operations Centre outside the MASS

4.41 Terms related to control: Control, Monitoring, Supervision (active/passive), strategic control, tactical control, supervisory control.

The following submissions have been proposed on these terms so far:

MSC 107/5/3

Operator control mode: This is a working mode, sometimes supported by technology or procedures, that represents the expected class of actions performed by the crew or remote-control centre operators. Modes can be changed during a voyage or operation and/or for specific functions. Four operator control modes have been defined as described in the following paragraphs.

Monitoring: An operator control mode with operations which monitor a situation but do not take any action to influence necessary processes. In monitoring mode, operators may adjust non-necessary processes or equipment to facilitate gathering of information. Monitoring can, for example, be to adjust a system for exclusively human use, such as external lights or cameras, or to inspect equipment or trends in performance parameters.

Strategic control: An operator control mode with operations to issue fleet-wide instructions that implement and, if appropriate, define specific functions to be used by the automatic decision-making units.

Tactical control: An operator control mode with operations to influence the conclusion made by the automatic decision-making units of the autonomous ship for a particular purpose. Tactical control includes, for example, changing the required minimum closest point of approach to other ships or the port of destination and letting the autonomous ship system afterwards construct the avoidance manoeuvre or route itself. It can also be adjustment of a technical alert level, based on prevailing conditions, for example, the time delay in actuation of the bilge alarm.

Direct control: An operator control mode with operations to control a specific function or parameter. Direct control means, for example, that the operator changes a waypoint that would otherwise be decided by the autonomous ship systems directly, or that the operator selects and overrides the machinery standby configuration, such as changing of generator or pump standby status.

RBAT (4th report)

Control: Purposeful action on or in a process to meet specified objectives (IEC, 2013).

Control function: Control actions performed by humans or software for the accomplishment of a functional goal (adapted from IEC, 2000).

Control action: Acquisition of information, analysis of information, decision-making, or implementation of physical actions performed as part of a control function.

Supervision: A role with an explicit responsibility to monitor system performance and detect abnormalities so that the desired outcome can be achieved through implementation of corrective responses

MSC 107/INF.8

Supervisory control – is a role with an explicit responsibility to monitor system performance and detect anomalies so that the desired outcome can be achieved through implementation of corrective responses. An important principle is that the supervisory agent cannot be the same as the agent performing the control action(s) being supervised. The supervisor has an overriding authority of the control action performance and is responsible for its outcome. Supervisory control can take different forms and be performed by either a software or human agent. The different categories of supervisory control defined in RBAT are:

Active human supervisory control - supervisor intervenes at any stage based on continuous monitoring

Passive human supervisory control - supervisor intervenes upon requests (e.g. alarm)

Software supervisory control - software intervenes on demand upon continuous monitoring of pre-defined parameters

4.42 Remote Operations – Term to be defined

4.43 *[Risk Assessment*

Risk Assessment means an assessment undertaken in line with/meeting the requirements of section 2.4 of this Code.]

[4.43bis Safe State – to be defined if used in the Code]

4.44 *[SOLAS*

SOLAS means the International Convention for the Safety of Life at Sea, 1974, as amended.]

4.45 Situational Awareness

[The classification of situational awareness capabilities should be categorized by mode of operation because the details of situational awareness will vary depending on the subject for which it is provided (crew, remote operators, and so on) and the functionality should differ. (MSC 107/5/7)

Situational Awareness means the perception of environmental elements and events with respect to time or space, the comprehension of their meaning, and the projection of their future status (Endsley 1995). (RBAT)]

4.46 Software - term to be defined if used in the Code

4.47 *[STCW Convention*

STCW Convention means the International Convention on Standards of Training, Certification and Watchkeeping for Seafarers, 1978, as amended.]

[4.47bis Submitter

Submitter is an entity seeking approval of a MASS from the Administration, responsible for communicating with the administration for the submission and follow up of the approval process.]

4.48 System

System means the combination of interacting elements functions organized to achieve one or more stated purposes, i.e., goals (IEC, 2018).

[4.48bis System Software - to be defined if used in the Code.]

[4.48ter Systematic failure events:

Systematic Failure means an event that is the consequence of inadequate work processes and may be introduced at all stages in the system lifecycle.]

[4.48quart Systemic failure

Systemic Failure means an event which occurs not only by the malfunctions of a distinct component in the system but due to inherent deficiencies flaws or oversights in the system's structure, including unforeseen interactions or dependencies among other various constitutive elements of the system.]

4.49 Task

Task means a set of [control] actions taken to enable functions and perform operations. A task may involve interactions with several different functions or systems, but also with humans.

or

[Task means a set of activities undertaken in order to achieve a specific goal. (ISO)]

4.50 Third parties

Third Parties means persons that are not involved in the operations but engaging with the MASS, e.g. VTS, ports, pilots or other persons in the ROC for maintenance reasons, persons in distress, other vessels.

4.51 [[Unannunciated]][Unannounced] failures

An *unannunciated failure* is one which is latent or fails 'passively', i.e., in such a manner as to not inform the crew of the failure by virtue of alerts, or the provided information is misleading, incomplete, or not presented in due time.]

4.52 Unsafe State

[Unsafe state means where a system is operating outside its operating envelope due to degraded performance (e.g., [faults or] failures) or exceeded capabilities which, if left [uncorrected or] unmitigated, has the potential to directly cause an accident.] (RBAT)

4.53 Verification

[Verification means the process of systematically evaluating and confirming that autonomous maritime vessels' design, technology, and operational protocols align with this Code and involves a thorough examination, also by testing, to ensure that the autonomous ship's construction, software systems, sensors, and control mechanisms comply with or meet the specific safety, navigation, and operational requirements set forth by this Code.]

4.54 Validation

[Validation means the process of thoroughly assessing and examining methodologies, assessments, procedures, hypotheses, or criteria used in the context of requirements or calculations.]

PART 2 MAIN PRINCIPLES FOR MASS AND MASS FUNCTIONS [AND REMOTE OPERATIONS]

CHAPTER 5 CERTIFICATE AND SURVEY

5.1 SOLAS Certificates

Any MASS should be furnished with appropriate SOLAS certificates. Any modification to SOLAS requirements due to the nature of the MASS should be reflected in the SOLAS certificates or by exemption certificates, but should also be reflected in the MASS certification. Any required records (e.g. to the SAFEQ Certificate) should be provided showing actual equipment on board.

The SOLAS certificates should be maintained as required by SOLAS and the Harmonized System of Survey and Certification, ref. Resolution A. 1156(32) as amended.

NOTE:

The MASS and ROC certificates are meant to certify the technical equipment of the ship and ROC including sensors, system integration and connectivity between the two (and others).

The MASS Certificate and MASS Record has been thought of from the perspective of a PSCO doing an expanded inspection – what do you need in order to ascertain whether the MASS is in compliance with the original approval? Additionally, the wording is kept as close as possible to current SOLAS wording (for better or worse).

5.2 MASS Certificate

Every ship to which this Code applies should have a valid MASS Certificate, issued after an initial or renewal survey.

5.2.1 The MASS functionality shall be subject to the following surveys:

- .1 an initial survey before the ship is put in service;
- .2 a renewal survey at intervals specified by the Administration but not exceeding five years;
- .3 a survey within three months before or after each anniversary date of the MASS Certificate; and
- .4 additional surveys as occasion arises.

5.2.2 The surveys referred to in 5.2.1 should be carried out as follows:

- .1 the initial survey should include verification and testing of the MASS functionality, to ensure that they comply with the requirements of this Code;
- .2 the renewal and periodical surveys should include verification and testing of the MASS functionality, to ensure that they comply with the requirements of this Code; and
- .3 the additional surveys either general or partial, according the circumstances should be made after a repair resulting from investigations prescribed in regulation 11 chapter I of SOLAS, or whenever any major repairs or renewals of equipment and/or systems related to autonomous or remote-control functions, to ensure that the MASS comply with the requirements of the

MASS Code, and of the laws, decrees, orders and regulations promulgated as a result thereof by the Administration. The major repairs or renewals of equipment and/or systems include but not limited to the following cases:

- a. Replacement of equipment and/or systems that may affect the compatibility and stability of the software system;
- b. Updated out of the OE;
- c. Repairs after serious failure or defects of equipment and/or systems;
- d. Other cases deemed necessary by the Administration.

5.2.3 The periodical surveys should be endorsed on the MASS Certificate.

5.2.4 The MASS Certificate should be accompanied by a MASS Record, containing the following:

- .1 a description of the MASS approved ConOps, OE, and MoO;
- .2 all operational restrictions on the MASS and the ROC (ref. 5.3.4);
- .3 survey requirements as associated with the system in question (ref. 5.3.4);
- .4 a list of all ROC approved for working with this particular MASS;
- .5 approved way of connection between MASS and controlling ROC; and
- .6 any exemptions from or equivalences to SOLAS or other IMO instruments made possible by employing the MASS Code.

The certificates and records of equipment should be drawn up in the form corresponding to the models given in appendix [NN] to this Code. If the language used is neither English nor French, the text should include a translation into one of these languages.

SOLAS Ch. I, reg. 6, 11, 13, 17, 19, 20 and 21 applies to the MASS Certificate.

5.3 MASS ROC Certificate

Every Remote Operation Centre (ROC) to which this Code applies should have a valid MASS ROC Certificate, for each [MASS/ship] it operates, issued after an initial or renewal survey.

5.3.1 The MASS ROC functionality should be subject to the surveys specified below:

- .1 an initial survey before the ROC is put in service;
- .2 a renewal survey at intervals specified by the Administration of the host nation but not exceeding five years;
- .3 a periodical survey within three months before or after each anniversary date of the MASS ROC Certificate; and
- .4 additional surveys, as occasion arises.

5.3.2 The surveys referred to in 5.3.1 should be carried out as follows:

- .1 the initial survey should include a complete verification and testing of the MASS functionality, to ensure that they comply with the requirements of this Code;

- .2 the renewal and periodical surveys should include verification and testing of the MASS functionality, to ensure that they comply with the requirements of this Code; and
- .3 the additional surveys either general or partial according the circumstances should be made after any major repairs or renewals of equipment and/or systems related to remote control functions, to ensure that they comply with the requirements of this Code.

5.3.3 The periodical surveys referred to in 5.1.1 should be endorsed on the MASS ROC Certificate.

5.3.4 The MASS ROC Certificate should be accompanied by a MASS ROC Record, containing the following:

- .1 a description of the ROC approved ConOps and OE;
- .2 all operational restrictions on the ROC (ref. 5.3.4).
- .3 survey requirements as associated with the system; in question (ref. 5.3.4);
- .4 a list of all MASS approved for working with this particular ROC;
- .5 approved way of connection between MASS and controlling ROC; and

Certificates and records of equipment should be drawn up in the form corresponding to the models given in appendix [NN] to this Code. If the language used is neither English nor French, the text shall include a translation into one of these languages.

SOLAS Ch. I, reg. 6, 11, 13, 17, 19, 20 and 21 applies to the MASS ROC Certificate, [with the condition that the Administration is to be understood as the Administration of the host nation of the ROC facility (physical location). The MASS ROC Certificate may cover national requirements of the host nation.]

NOTE:

The ISM Certification for MASS are meant to certify the operational requirements of the MASS, it's responsible Company and the ROC[s] in question.

This will have to be further developed in the light of document MSC 108/4/2 and others.

5.4 Safety Management Certification

Every ISM Company [intend to operate a ship with autonomous or remotely controlled functions]/[conducting autonomous or remotely controlled operations] should develop its Safety Management System to address MASS Operations, including Remote Operations, and [tasks] assigned to a ROC [when applicable].

The ISM Company should ensure that the SMS is implemented and maintained on the ship and at any ROCs involved in its operation. The ISM Company should clearly identify and document the subdivision of [tasks] and relationship between the ISM Company and any associated ROCs [and ships].

Any operational procedures specified by this code for the autonomous or remotely controlled operations, including watchkeeping arrangements, should be included in the safety management system of the ship and /or ROC respectively.

The operation of the ROC should, to the satisfaction of the Administration, be included in the ISM verification and certification process for an ISM company.

5.4bis Alternatively, a Company operating Remote Operation Centres [ROM Company], to the satisfaction of the Administration, could establish a Safety Management System for the ROCs under their responsibility. This ROC SMS (ROCMS) should be used in conjunction with the ship's SMS issued by the ISM Company and a clear link [and responsibility] between the two should be established, [ensuring, in any case, that a similar safety level to that established through ship's SMS is maintained].

[The ROM certification should follow the guidance laid out in *(to be developed)*, with-]

[The alternative ROM certification should follow a similar approach as the ISM Code with]

- a ROC Safety Management System (ROCMS)
- a DoC for the ROM Company (ROM DoC)
- a ROC Management Certificate (RMC) for each ROC
- a MASS Certificate Remote Operations Supplement for the ship together with a MASS ROC Certificate for the ROC to establish the link between ROC and ship

5.5 ISPS Certification for MASS

Any MASS should be ISPS Certified according to the ISPS Code. The ISPS procedures may be integrated with the ISM system, as long as the required confidentiality is observed.

If the ISM and ISPS systems are integrated, the approval process for ISPS [will] [should] follow the procedures in the ISM Code and not the ISPS Code procedures. *Note, this is a deviation from standard ISPS procedures and need to be discussed.*

5.6 Minimum Safe Manning Documents

Both the MASS and any ROC operating a MASS shall be furnished with a Minimum Safe Manning Document (MSMD) according to IMO resolution A.1047(27) as amended, and to the satisfaction of the Administration.

The MSMD for the MASS may carry a total manning number of 0 (zero) and may reference personnel training and certification requirements as specified in this Code.

The MSMD for the ROC should be linked to specific or a number of specific MASS. As the STCW Convention and Code does not apply to ROC, the MSMD for the ROC may include personnel outside of STCW and may reference personnel training and certification requirements as specified in this Code.

The ROC may employ a number of MSMDs dependent on the needs of the individual MASS operated by the ROC. If the ROC operates under more than one MSMD, a MASS ROC Master Plan (MRMP) for watchkeeping and other tasks has to be approved by all Administrations who has issued MSMDs covered by the MRMP. The MRMP may be part of an ISM System.

NOTE:

If the ship is operated from a ROC, with or without crew on board, a PSCO will need access to the MSMD for the ROC in order to ascertain that the ship is adequately manned. This needs further discussions.

Footnotes

⁴ Refer to paragraph 4.4.3 of resolution A.1118(30) – Revised Guidelines on the implementation of the ISM Code by Administrations

⁵ Refer to paragraph 16.2 of part B of the ISM Code

CHAPTER 6 APPROVAL PROCESS

6.1 Process description

A structured approval process should take place to enable the MASS to obtain the required approval including the necessary certificates related to requirements for the intended operation. By following this process, Submitters and Administrations would be working in cooperation to ensure that all aspects of safety, security and environmental protection are adequately assessed. This process is intended for the MASS as a whole, while individual systems are covered in the verification and validation step.

The approval process for MASS should be based on and follow the main principles of the *Guidelines for the approval of alternatives and equivalents as provided for in various IMO instruments* (MSC.1/Circ.1455) taking into consideration parts 2 and 3 of this Code. The level of detail should be proportional to the complexity, level of novelty and associated risk of the MASS and on whether the Submitter is applying for preliminary or final approval and the necessary documentation may vary accordingly.

The steps and documentation required in this chapter provide the general basis of the approval process. This does not exclude other information or documentation that may be requested by the Administration or an alternative process which may be followed to the satisfaction of the Administration.

Sufficient information should be supplied to enable the Administration to fully assess the features of the MASS. After appropriate identification of relevant stakeholders by the Submitter, discussions should commence at the earliest possible stage so that the Administration may fully evaluate the level of safety of the MASS.

The approval process should be conducted using the following steps:

1. Preliminary design development;
2. Preliminary design approval;
3. Testing, simulation, and other verification methods;
4. Final approval;
5. [Survey and Certification]; and
6. Operation.

6.2 Evaluation criteria

The basic principle for the evaluation criteria should be to ensure a level of safety, environmental protection and security that is expected of a conventionally operated ship. The evaluation criteria should be developed through compliance with the goals and functional requirements of part 3 of this Code in combination with a risk assessment (as described in 7.2 of the Code). The evaluation criteria and an assessment plan thereof should be agreed with the Administration.

6.3 Design and documentation requirements

For each approval step, information and documentation required by the Administration should be produced and submitted. The various documents required in the approval process steps are expected to be reviewed according to any possible design or operational changes and added details. The approval steps do not need to be sequential, meaning that they may also run at the same time.

The ConOps (as described in section 8.2) should be a base document in the approval process and should be the basis for the assessment in each step.

[6.3.4 Survey and certification

Survey and certification requirements as described in 5.8 of the Code should be followed.]

6.3.5 Operation

Conditions for maintaining the safety level agreed during the design approval may be imposed on ship operation. Any operational conditions should be determined during the approval process, and they should be clearly documented and communicated to relevant parties.

If, during the operational phase, the initial assumptions, systems and equipment are changed, i.e. any change in the ConOps, the part of the risk assessment with the respective changes should be repeated. The extent of work needed will be dependent on the risk-based features, the changes and the operation of the ship and may be decided by the Administration. For example, in the case of a ROC with remote operators approved for the control of one vessel, it may be necessary to review the approval assumptions if a second vessel is added to the same ROC.

MASS may initiate operations while it is being tested. During such tests, the certification of the ship should follow [the interim certification process described in 1.6. In addition, further mitigation layers to the system that is being tested should be foreseen such as different modes of operation from the ones of the final design.

Table 1 in the annex provides additional guidance in relation to the approval steps that require relevant documentation:

CHAPTER 7 RISK ASSESSMENT

7.1 A risk assessment should be conducted to ensure that risks arising from the use of MASS functions, including relevant functions in ROCs, affecting persons on board, the environment, and the safety of the ship are addressed, taking into account identified goals and functional requirements, ensuring a level of safety expected of a conventional ship. The risk assessment can be conducted on MASS as a whole, and/or on the MASS functions. It should also consider the ConOps (and its OE) of the MASS. The risk assessment should address relevant mitigation measures. Should the risk assessment be carried out on specific MASS functions, the consequences on other ship's functions should be considered and mitigated.

The risk assessment should be used for the approval process as required in chapter 6 (approval process).

Proposal by Norway (MSC/ISWG/MASS 3/3) - NOTE: Proposal not discussed at MSC/ISWG/MASS 3 – included for ease of future reference.:

7.2 ~~The Appropriate~~ risk assessment methodologies should be used for the different steps of the approval process as required in chapter 6 (approval process). Such risk assessments might include but are not limited to

- .1** MASS (including ROCs) and system design phase;
- .2** alterations and modifications of a major characteristics of the ship or of the OE or ConOps of MASS that may have impacts on MASS functions.

7.2bis A risk assessment should be carried out by personnel with relevant expertise as required by the Administration of the flag State (MSC.1/Circ.1212/Rev.1, annex point 4). A Risk assessment may be performed at the following stage, including but not limited to:

- .1** MASS (including ROCs) and system design phase (refer to paragraph 1 of the [annex](#) for preliminary design development requirements);
- .2** after alterations and modifications of a major characteristics of the ship or of the OE or ConOps of MASS that may have impacts on MASS functions (refer to paragraph 6.3.5 for operation phase requirements).

Proposal by Norway (MSC/ISWG/MASS 3/3) - NOTE: Proposal not discussed at MSC/ISWG/MASS 3 – included for ease of future reference.:

7.2bis A risk assessment should be carried out by personnel with relevant expertise as required by the Administration of the flag State (MSC.1/Circ.1212/Rev.1, annex point 4). ~~A Risk assessment may be performed at the following stage, including but not limited to:~~

- ~~**.1** MASS (including ROCs) and system design phase (refer to paragraph 1 of the [annex](#) for preliminary design development requirements);~~
- ~~**.2** after alterations and modifications of a major characteristics of the ship or of the OE or ConOps of MASS that may have impacts on MASS functions (refer to paragraph 6.3.5 for operation phase requirements).~~

7.3 Risks should be analysed using suitable, recognized and appropriate risk assessment techniques.¹ The output format should be justified by the Submitter and be agreed between the Submitter and the Administration. Risk assessment should include a comprehensive description of the autonomous and remote-control function's utilization, effectiveness and reliability performing a thorough hazard analysis, conducting a mitigation [corrections and] analysis, evaluating the identified risks, and implementing effective risk control measures. The risk assessment should analyse and address hazards associated with the intended OE of the MASS including the associated ROCs, as described in the ConOps. Apart from the hazards such as loss of function, cyber attacks, component damage, fire, explosion and electric shock, it should also consider the random, systematic, and systemic hazards involved within the OE (refer to paragraph 2 of the [annex](#) for preliminary design approval steps related to risk assessment report).

Proposal by Norway (MSC/ISWG/MASS 3/3) - NOTE: Proposal not discussed at MSC/ISWG/MASS 3 – included for ease of future reference:

7.43 Risks should be analysed using suitable, recognized and appropriate risk assessment techniques.¹ The output format should be justified by the Submitter and be agreed between the Submitter and the Administration. Risk assessment should include a comprehensive description of the autonomous and remote-control function's utilization, effectiveness and reliability performing a thorough hazard analysis, conducting a and mitigation [corrections and] analysis, evaluating the identified risks, and implementing effective risk control measures. The risk assessment should analyse and address hazards associated with the intended OE of the MASS including the associated ROCs, as described in the ConOps. Apart from the hazards such as loss of function, cyber attacks, component damage, fire, explosion and electric shock, it should also consider the random, systematic, and systemic hazards involved within the OE (refer to paragraph 2 of the annex for preliminary design approval steps related to risk assessment report).

7.4 The adopted mitigation measures should take into consideration single failure events, but also foreseeable events within the OE of the ship that may influence the performance of more than one system at the same time (e.g. heavy weather during hours of darkness). Such features should consist mainly of independent mitigation layers, including predefined fallback states. The number of such mitigation layers should be proportional to the risk. The assessment should ensure that hazards are eliminated wherever possible through inherently safe design and hazards that cannot be eliminated should be mitigated as needed. The assessment should ensure that hazards are eliminated wherever possible through inherently safe design and hazards that cannot be eliminated should be mitigated as needed, with the details of hazards and the means of mitigating them being documented to the satisfaction of the Administration. The effectiveness of the mitigation measures adopted in the risk assessment should be verified according to the verification and validation plan stipulated in refer to paragraph 2.7 of the annex.

Proposal by Norway (MSC/ISWG/MASS 3/3) - NOTE: Proposal not discussed at MSC/ISWG/MASS 3 – included for ease of future reference::

7.54 The adopted mitigation measures should take into consideration single failure events, but also foreseeable events within the OE of the ship that may influence the performance of more than one system at the same time (e.g. heavy weather during hours of darkness). Such features should consist mainly of independent mitigation layers, including predefined fallback states. The number of such mitigation layers should be proportional to the risk. The assessment should ensure that hazards are eliminated wherever possible through inherently safe design and hazards that cannot be eliminated should be mitigated as needed. The assessment should ensure that hazards are eliminated wherever possible through inherently safe design and hazards that cannot be eliminated should be mitigated as needed, with the details of hazards and the means of mitigating them being documented to the satisfaction of the Administration. The effectiveness of the mitigation measures considered adopted in the risk assessment should be verified according to the verification and validation plan stipulated in refer to paragraph 2.7 of the annex.

¹ Refer to MSC.1/Circ.1455 and IEC/ISO 31010:2019 – Risk assessment techniques and Risk assessment Methodologies may be used include: IEC 61508 Parts 1 to 7 – Functional safety of electrical/electronic/programmable electronic safety;

STAMP: http://psas.scripts.mit.edu/home/get_file.php?name=STPA_handbook.pdf

RBAT : <https://www.emsa.europa.eu/mass/rbat.html>

CHAPTER 8 OPERATIONAL CONTEXT

8.1 The operational context for a ship should, within the applicable regulatory framework, and on the assumption that it does not have any privilege over conventional ships, consider all aspects of the MASS operation and describe the autonomous or remotely operated ship function(s) and the external environment that influences its operation.

The operational context encompasses the Concept of Operation (ConOps), the Operational Envelope (OE), the system-specific Operational Design Domain (ODD), fallback states, the Mode(s) of Operation (MoO) and the operator control modes (OCM).

The ConOps, OE, ODD, fallback state and MoO should be part of the certification. (*N.B. Final location to be confirmed*)

8.2 Concept of Operation (ConOps)

8.2.1 The ConOps should be drafted to ensure the safe, secure and environmentally friendly operation of the ship. Risk assessments for the ship and ROC should take the ConOps into consideration. The ConOps and the associated risk assessment should ensure that all relevant risks are addressed.

8.2.2 The ConOps should include consideration of the Operational Envelope (OE) and the technical design of the ship and of the Remote Operation Centre(s) (ROC), if applicable, including the connectivity and communication lines. The ConOps should address the control, monitoring, and intervention on board the ship and at the ROC, if applicable, together with the integration of humans in the operation. The ConOps should be reviewed as and when there are hardware, software, operational and management changes to the ship or ROC.

8.2.3 The ConOps should be reviewed when there are hardware, software, operational and management changes to the ship or ROC.

8.3 Operational Envelope (OE) and Operational Design Domain (ODD)

The OE of the ship as an integrated system should incorporate the ODD of all individual autonomous or remotely operated systems or functions.

8.3.1 Operational Envelope (OE)

The Operational Envelope (OE) of the ship should encompass the operational capabilities and limitations of the autonomous or remote operation, and ship-specific capabilities and limitations to indicate the condition in which an autonomous or remotely operated ship can operate safely in all operating conditions, including all reasonably foreseeable degraded states.

The OE should contain:

- .1 the definition of the ship functions and conditions and its use case(s);
- .2 the geographic area of operations, including coverage/connectivity and traffic conditions;
- .3 the description of the environmental limitations;
- .4 the description of operational limitations for different modes of operation during a single voyage;
- .5 the use and management of the modes of operation, including the division of functions and allocation of tasks between humans and automation; and
- .6 any other factors that have a significant impact on MASS operations.

8.3.2 Operational Design Domain (ODD)

The Operational Design Domain (ODD) of individual functions or systems, should be based on the results of a risk assessment. The relevant design documents should include:

- .1 the conditions and limitations under which any individual autonomous or remotely operated system or function operates safely, including all reasonably foreseeable degraded states;
- .2 the extent of human interaction;
- .3 the capabilities and limitations to be accomplished before activation of operation of the individual system or function;
- .4 descriptions of the external and internal conditions, such as geographical boundaries the ship is to operate in, the maximum wind and sea wave heights etc.; and
- .5 reasonably foreseen system or equipment malfunctions.

An autonomous or remotely operated system and related equipment should operate within its ODD and should be able to detect whether its current state of operation meets the ODD.

8.3.3 Degraded state

A single autonomous or remotely operated system or function deviating from its ODD should not necessarily result in the ship deviating from its OE. As long as the ship as an integrated system can continue to be operated within its OE the deviation of an autonomous or remotely operated system or function from its ODD should be considered as a degraded state.

8.4 Fallback state

In case of deviating from its OE a ship should enter a predefined fallback state, offering an additional mitigation layer. There may be more than one predefined fallback state to address failures or conditions identified at the ship design stage that might lead to a ship deviating from its OE.

When a ship enters a fallback state, the predefined fallback action(s) should avoid, as far as practicable, any harm to life at sea, other ships, infrastructure, or the marine environment until the ship returns into its OE and the normal operation is restored.

Fallback states should be risk-assessed in order to demonstrate effectiveness to avoid further deterioration in the status of the ship and increasing the threat to life at sea, to other ships, infrastructure, or the marine environment. Depending on the result of the risk assessment, more than one independent fallback state should be available at any time during normal operations. Being in a fallback state should not result in an intolerable risk.

The actions and procedure to enter into and recover from a fallback state should be considered as the fallback response and should be predefined [and demonstrable].

The ship should notify its crew and the operator when transitioning to, and operating in, a fallback state.

8.5 Contingency plans

A deviation from an OE beyond a predefined fallback state, should lead to the activation of contingency plan(s).

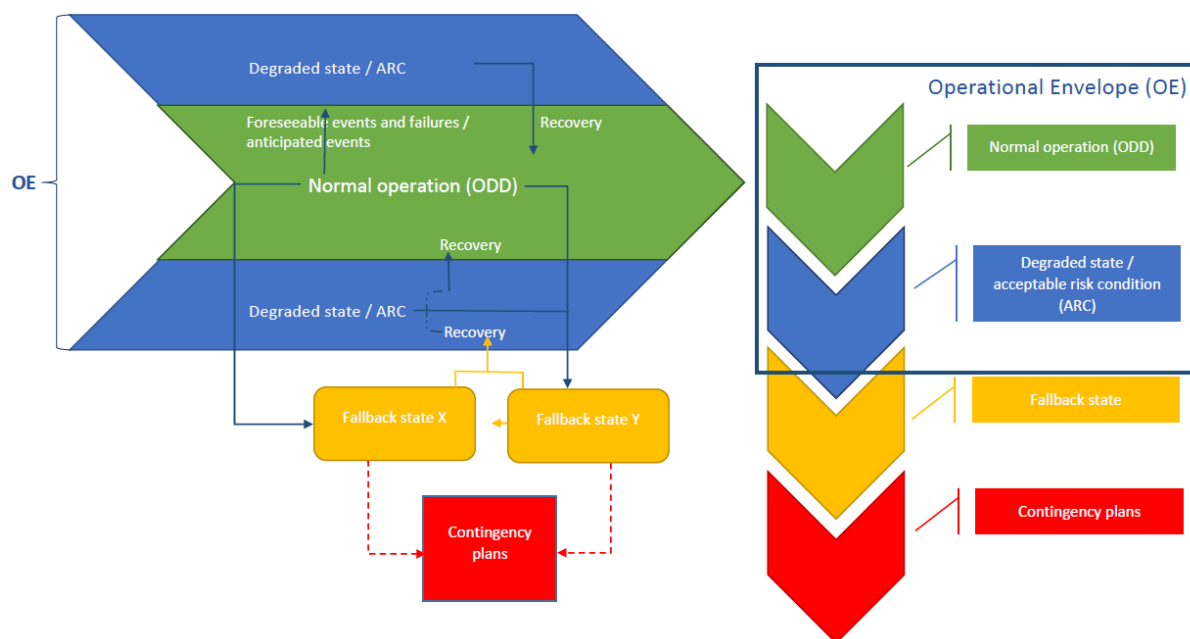


Figure 1: Illustration OE – ODD – degraded state – fallback

8.5 Mode(s) of Operation (MoO)

The MoO of a ship may be changed for different phases of a voyage and procedures to change from one MoO to another, along with the criteria for any such change, should be described in the ConOps.

The description of MoO should also identify:

- [.1 which ship functions are autonomous or remotely operated;]
- .2 how autonomous or remotely operated ship functions are allocated to different agents (human or software);
- .3 how the affected ship functions are supervised, and by which agents;
- .4 where the different agents are located (on board or remote); and
- .5 which other systems and other roles (personnel) are involved in performing the control action.

[8.6 Operator Control Modes (OCM)

Identification of other systems and other roles (personnel) involved in performing control actions as per 1.5.5 should be based on the understanding that the MoO of a ship without any human interaction is limited and, as a last resort, it should be foreseen that humans are able to exercise control, should the situation require it. The Operator Control Modes should provide remote or onboard crew ample time to exercise direct or supervisory control of all functions according to the [defined][approved] ConOps.

As a minimum to account for assuming responsibility, the ship remote or onboard crew should have sufficient input to obtain situational awareness, and means to control the ship. Both should be available in ample time to perform an appropriate control action in any situation.]

CHAPTER 9 SYSTEM DESIGN

In addition to complying with relevant rules and regulations and standards, performing and supervising any specific function of the ship, MASS functions should comply with the following high-level principles.

9.1 Safety-Centric Design:

Systems should be designed to minimize risks to the ship, crew, ROC operators, cargo, other ships and the marine environment by incorporating inherently safe design principles. All systems used for MASS operations should include fail-safe mechanisms and emergency protocols to ensure comprehensive safety and effective risk management. Hazards affecting the systems should be eliminated wherever possible, and those that cannot be eliminated should be mitigated as needed by using a risk assessment as described in chapter 7.

9.2 Human Centered Design

[The design and operation of the MASS and associated (ROC) human-machine interfaces should prioritize assistance to human operators. The Human-Centered Design (HCD) should be appropriately tailored for all anticipated interactions between the crew/operators and the MASS.

Interactions between onboard crew and remote operators should be considered in the Human-Centered Design outlined in section 13.6.

Human operators, both on board and in remote, should be actively involved in the validation stages to ensure that the systems can be safely operated.]

9.3 Robustness and Reliability:

- .1 Systems should be robust and should be able to operate effectively under adverse conditions, including diverse maritime environments and operational challenges.
- .2 It should be ensured that the systems perform their required functions effectively during the operational period specified by the manufacturer, up to predetermined maintenance intervals.

9.4 Adaptability and Flexibility:

Systems should have the ability to adapt to changing environments, tasks, and user requirements, and allow for updates and modifications to accommodate necessary technical and regulatory updates, and future needs.

9.5 Redundancy and Fault Tolerance

- .1 redundant sub-systems should be implemented to maintain functionality in case of component failures including systemic or systematic failures.
- .2 Systems should be designed to handle and recover from failures and continue operating at a reduced performance level (fall back state).

9.6 Scalability:

It should be ensured that systems design are scalable, allowing for expansion or updates as technology advances or operational needs change.

9.7 Security and Cybersecurity:

Security measures to protect the systems on the MASS and the ROC should be incorporated to prevent unauthorized access and cyber threats.

9.8 Data Management and Quality:

Efficient data management systems should be incorporated to ensure data accuracy, integrity, and quality [and design systems to leverage data for enhanced performance and decision-making].

9.9 Interoperability:

Compatibility and interoperability with systems, devices, applications, and technologies should be ensured.

9.10 Testing and Validation:

MASS systems should undergo comprehensive testing and validation to ensure compliance with design specifications and operational requirements. This process includes a structured, procedure comprising detailed simulation, component testing, integration testing, system testing.

Operators should be actively involved in the system validation phase in real-case scenarios to ensure practical usability and operational soundness meeting all regulatory requirements, before obtaining the necessary certifications.

9.11 Transparent Design:

Ensure that transparency is maintained in the system design for systems operations and decision-making processes.

CHAPTER 10 SOFTWARE PRINCIPLES

The following principles should be implemented to ensure that software on or supporting MASS (or automated functions thereof) are [reliable] trustworthy, safe and secure. They should be used within the context of complying with the MASS Code, including the use of remote control and autonomous operation of key functions.

The principles should be considered as part of the approval process, and this may be done using software quality assurance standards*.

10.1. Proportionality

Software should have an explicit and well-defined operational design domain. The use of software should not go beyond what is provided for in the ConOps and risk assessment(s) should be used to prevent hazards which may result from such uses.

-
- MSC.1/CIRC.1512 *Guidelines on Software Quality Assurance and Human-Centred Design for E-Navigation*
 - IEC 61 508 Functional safety of safety-related electrical/electronic/programmable electronic systems
 - INS standard (IEC 61924-2),
 - IEC 61162 series of standards
 - IACS UR E22 "Computer-based systems". The Rev.3 (June 2023)
 - ISO/IEC 90003 Software engineering - Guidelines for the application of ISO 9001 to computer software [ISO/IEC 25000]
 - MSC-FAL.1/Circ.3/Rev.2 *Guidelines on Maritime Cyber Risk Management*

[10.x Reliable

The effectiveness of such software capabilities should be subject to testing and assurance within the ODD across their entire lifecycles.]

10.2. Safety and Security

Unwanted harm (safety risks) as well as vulnerabilities to external factors (security risks) should be avoided and addressed. Safety and security (including cybersecurity) risks should be identified, addressed, and mitigated throughout the software's operational life to prevent and/or limit, any potential or actual harm to shipping, humans, or the environment.

10.3. Transparency and Explainability

Software should be transparent and explainable at all stages of its operational life, and for all decision-making processes. The transparency and explainability should ensure relevant personnel possess an appropriate understanding of the technology, development processes, and operational methods applicable to autonomous capabilities, including with transparent and auditable methodologies, data sources, and design procedure and documentation.

10.4. Accountability

[Mechanisms should be implemented to provide accountability over the Organizations and individuals developing, deploying, or operating software to ensure proper operation.] Software should be auditable and traceable to such organizations and individuals. There should be [governance] mechanisms in place for oversight, impact assessment, audit, and due diligence to ensure accountability for the software's impact throughout its operational life.

[10.x Governable

Software should be designed and engineered to fulfil their intended functions while possessing the ability to detect and avoid unintended consequences, and the ability to disengage or deactivate deployed systems that demonstrate unintended behaviour.]

10.5. Robustness

Safe and secure software and hardware should be enabled through robust frameworks. Software should perform consistently with intended objectives, in a stable and resilient manner in a variety of circumstances. [Processes should be in place for managing continual improvement of software and provide for adapting situations.] The robustness of such systems should be tested and assured across their entire life cycle within that domain of use.

10.6. Human Oversight and Determination

Software should be designed and developed to ensure people managing MASS operations can exercise [meaningful] oversight, including the ability to verify decisions when required. Humans should be provided with the means to interpret appropriate context, prevent or minimize risks, and contest decisions that impact the safe, secure, and environmental sound operation of MASS.

[An audit log functionality in line with the principles of this chapter should be established on the MASS that is readily accessible for review and analysis on board and at the ROC. The log should be retained on board and at the ROC for a sufficient period of time to support navigational safety and casualty investigation. The retention period should be in accordance with the requirements of the Administration. *N.B. CAN, by email, 17-May-24*][*N.B.this par. should be relocated to another chapter*]

[10. x Unintended Bias

Software should be designed and developed to prevent unintended bias. [Appropriate actions must be taken to mitigate systemic system bias or discriminatory outcomes for individuals and groups to avoid unintended bias.]]

CHAPTER 11 MANAGEMENT OF SAFE OPERATIONS

11.1 Goal

The goal of this chapter is to ensure adequate management for safe operations.

11.2 Functional Requirements

To achieve the above-mentioned goal, the ship should comply with the requirements for the management for the safe operation of ships in SOLAS and the ISM Code as modified and/or supplemented by the functional requirements of this chapter.

11.2.1 The Safety Management System (SMS) should provide for safe MASS operations by establishment of procedures, plans and instructions for all foreseeable modes of operation, including those involving different physical locations, if applicable.

- EP 1 Risk control measures should address autonomous or remotely controlled ship functions.
- EP 2 Internal processes verifying the effectiveness of procedures, plans and instructions should address autonomous or remote-controlled ship functions.
- EP 3 Human-machine-interface aspects of autonomous or remote-controlled ship functions.
- EP 4 role and expected performance for all physical location(s) where autonomous or remote-controlled ship functions are executed.
- EP 5 With respect to autonomous or remote-controlled ship functions, the following should be considered:
 - .1 their interaction, capabilities and limitations;
 - .2 the complexity of systems, including software systems or data services;
 - .3 equipment and systems necessary to maintain contact to the MASS.
 - .4 lines of communication to maintain contact to the MASS;
 - .5 cyber-threats; and
 - .6 fall-back actions and processes to maintain their safe operation

11.2.2 The Safety Management System (SMS) of the company should provide for the safety and well-being of the personnel involved in the operations by:

- .1 identification of resources and training required; and
- .2 establishment of procedures, plans and instructions for all foreseeable operating conditions of the ship, including those involving different physical locations, if applicable.

- EP 1 This will be accomplished by ensuring consideration of:
- .1 risk control measures addressing autonomous or remote-controlled ship functions;
 - .2 internal processes verifying the effectiveness of procedures, plans and instructions addressing autonomous or remote-controlled ship functions;
 - .3 human-machine-interface aspects of autonomous or remote-controlled ship functions;
 - .4 responsibilities with regard to the [intersection and] interaction in MASS operations;
 - .5 how to maintain function of overriding authority; and
 - .6 emotional pressure, specific stresses and strains to humans in the MASS operations.

11.2.3 The Safety Management System (SMS) of the company should provide for the safety of the ship under all expected emergency conditions by establishment of contingency procedures, plans and instructions, including emergency scenarios involving different physical locations, if applicable.

- EP 1 This will be accomplished by ensuring consideration of:
- .1 risk control measures addressing autonomous or remote-controlled ship functions;
 - .2 internal processes verifying the effectiveness of procedures, plans and instructions addressing autonomous or remote-controlled ship functions;
 - .3 human-machine-interface aspects of autonomous or remote-controlled ship functions;
 - .4 monitoring autonomous or remote-controlled ship functions performance including relevant system and ship parameters;
 - .5 assistance for emergency handling or handling of other potentially unsafe situations; and
 - .6 capabilities and limitations of emergency response in the MASS operation.

CHAPTER 12 CONNECTIVITY

12.1 [Appropriate] connectivity should be ensured [at all times] of MASS operations, including between the MASS and ROC. To achieve this, redundancy measures should be implemented based on the risk assessment.

Proposal by China (MSC/ISWG/MASS 3/3/4) - NOTE: Proposal not discussed at MSC/ISWG/MASS 3 – included for ease of future reference::

12.1 [Appropriate] Connectivity should be ensured ~~[at all times]~~ of MASS operations [according to the ConOps/OE of MASS], including between the MASS and ROC. ~~To achieve this, redundancy measures should be implemented based on the risk assessment.~~

12.2 The infrastructure for connectivity, and its performance, should be acceptable to by the Flag Administration responsible for the MASS or a Recognized Organization.

12.3 The connectivity between MASS and ROC should be established using redundancy measures, including main and backup measures, preferably using different connectivity technologies [, bandwidth]/[, frequency] or/and service providers, as necessary according to the ConOps.

Proposal by China (MSC/ISWG/MASS 3/3/4) - NOTE: Proposal not discussed at MSC/ISWG/MASS 3 – included for ease of future reference:

12.3 ~~The connectivity between MASS and ROC should be established using r~~Redundancy measures should be implemented to establish connectivity between MASS and ROC based on the risk assessment, including main and backup measures, preferably using different connectivity technologies [, bandwidth]/[, frequency] or/and service providers, as necessary according to the ConOps.

12.4 The connectivity between MASS and ROC should be operated according to appropriate Quality of Service (QoS) requirements.

Proposal by China (MSC/ISWG/MASS 3/3/4) - NOTE: Proposal not discussed at MSC/ISWG/MASS 3 – included for ease of future reference:

12.4 The connectivity between MASS and ROC should be operated according to appropriate Quality of Service (QoS) requirements ~~which includes but not limited to the use of the minimal agreed, or better levels of acceptable latency and bandwidth.~~

12.5 The connectivity [connections] should be such as to operate the MASS safely using the minimal agreed levels of acceptable latency and bandwidth, or better. It should also consider the operational limitations, such as meteorological/oceanographic conditions (fog, wind, rain, thunderstorm, swell, etc.)

Proposal by China (MSC/ISWG/MASS 3/3/4) - NOTE: Proposal not discussed at MSC/ISWG/MASS 3 – included for ease of future reference::

12.5 The connectivity [connections] should be such as to operate the MASS safely. ~~using the minimal agreed levels of acceptable latency and bandwidth, or better. It should also consider the o~~Operational limitations of connectivity, such as meteorological/oceanographic conditions (fog, wind, rain, thunderstorm, swell, etc.) ~~should also be considered to ensure the safe operation of MASS.~~

12.6 The connectivity between MASS and ROC should continue operating at full capacity even in the case of a single failure in the system for realizing the connectivity.

12.7 The data exchanged during in the connectivity with the ROC should be categorized and prioritized according to a pre-defined prioritization scheme to enable data with higher priority to prevail on lower prioritized data in case of decrease in communication capacity. The pre-defined categorization and prioritization of exchange of data should be included in the ConOps.

12.8 The connectivity should be monitored for real-time or near real-time against its performance requirements. If disconnection or performance degradation of the main connection is detected, the system should automatically switchover to a backup connection through means described in the Operational Envelope of the ship. If both, the main and the backup connections do not meet the connectivity requirements, the MASS should enter a Fallback state with alert to ROC until the connectivity automatically reestablished. Measures should be taken to alert stakeholders to any disconnection issues and recover from the abnormal condition. Automatic reestablishment should be in accordance with the specifications within the OE of the ship.

12.9 Connectivity including Computer Based System (CBS)* onboard MASS and ROCs should ensure the integrity of transmitted data. At the same time, measures** should be taken to protect the security of transmitted data. [To prevent unauthorized access and interference]

* Refer to IACS UR E26 [and UR 22/Rev.3 and UR E27].

** Refer to MSC.1/Circ.1639 and MSC-FAL.1/Circ.3/Rev. 2.

CHAPTER 13 RADIOCOMMUNICATIONS

13.1 Goal

The goal of this chapter is to ensure adequate communication with all relevant entities (ships, ROCs etc.)

13.2 Functional requirements

To achieve the above-mentioned goal, a ship should comply with all relevant radiocommunication requirements of SOLAS, as modified and/or supplemented by the functional requirements of this chapter.

13.2.1 Measures should be taken to [ensure] [establish] that the communication between the ship and all relevant entities is achieved.

- EP 1 The technologies used in communication should not be limited to those used in GMDSS.
- EP 2 The communication should consider cybersecurity.
- EP 3 Communication should be available during the voyage.
- EP 4 The communication should be reliable, stable, and secure.
- EP 5 The communication should at all times operate with a quality of service ensuring, in particular, sufficient bandwidth and minimal transmission errors and latency, to support the necessary interaction between the ship and all relevant entities. This minimum quality of service should be maintained during the voyage.

13.2.2 The communication status of the ship should be monitored and an alarm with sound and visual indication should be provided to relevant operators on board or in ROCs in the event of communication failure. Details of communication failures and response means should be reflected in the ConOps.

13.2.3 An ROC should be able to communicate with one or more ships.

EP 1 When ROC communicates with more than one ship simultaneously, these communications should not affect each other.

[EP 2 If applicable, ROC should seamlessly switch to communication with other MASS supposed to be under its control when her communication with previous ROC.]

13.2.4 Where remotely operated, a ship should be able to communicate with one or more ROCs.

EP 1 The ship should be controlled by only one ROC at any given time.

EP 2 If a ship can be operated by multiple ROCs, communications between the ROCs should not endanger safe operation of the ship.

13.2.5 For remotely operated ships, communication between the ship and the ROC should be with the ROC operators.

EP 1 The ship should automatically transmit received external information to its ROC without delay.

13.2.6 Means should be provided for ROC to communicate with automated systems on the ship.

EP 1 The ship should transmit and receive relevant information and communicate with the ROC.

EP 2 Distress alerts of the ship must be automatically generated on board and transmitted when required and false alerts are to be avoided*.

* see resolution MSC.514(105)

13.2.7 The ROC should be capable of continuously monitoring all operational aspects of the ship, including but not limited to the OE and the mode of operation.

13.2.8 [Regardless of the manufacturer,] Compatibility and communication between different ROCs should be ensured to facilitate safe and efficient interoperation among ships.

CHAPTER 14 ALERT MANAGEMENT

14.1 Goal

The goal of alert management is to enhance [and optimize] the monitoring, handling, distribution, and presentation of alerts for MASS and ROC [during normal operation, fallback states, and emergency situations] [facilitating operators' focused attention on ensuring the safe operation while promptly addressing any alert situation that arises.]

14.2 Functional Requirements

To achieve the above-mentioned goal, the following functional requirements which are supplementary to SOLAS requirements are embodied in this chapter.

FR 1 An alert management optimization should be performed taking into account the ConOps so that the alert management provides:

- .1 the means used to draw the attention of the remote and onboard crew to the existence of alert situations;
- .2 the means to enable the human operator to identify and [address] [evaluate] that condition;
- .3 the means for MASS remote and onboard crew [and third parties] to assess the urgency of different abnormal situations in cases where more than one abnormal situation has to be handled;
- .4 the means to enable a human operator to handle alert announcements;
- .5 the means to manage all alert related states in a distributed system structure in consistent manner; and
- .6 the means for effective supervision of the autonomous and remote-control functions

14.3 Expected Performance

[EP 1 Alerting should follow the basic principles of the Bridge Alert Management (BAM) concept outlined by MSC.302(87) [and the 2009 Alert & Indicator Code (A.1021(26))]. Alert prioritization should reflect the urgency for action by the operator taking into consideration chapter 6 of MSC.302(87).]EP 1bis If practicable, there should be no more than one alert [per operator] for one situation that requires attention.

EP 2 Alerts should ideally provide sufficient information for the operator to grasp the consequences of the situation and determine appropriate actions. However, it's recognized that in some cases, brevity may preclude detailed comprehension.

EP 2bis The alert management should [handle alerts from all systems or components required to support MASS and ROC operation in accordance with] [be able to handle all alerts required by] performance standards adopted by the Organization. Furthermore, it should have the capability to handle all other alerts in a consistent manner.

EP 3 The logical architecture of the alert management and the handling concept for alerts should provide the capability to minimize the number of alerts especially those on a high priority level (e.g. using system knowledge from redundancy concepts [inside the ANS] and evaluating inherent necessities for alerts against navigational situations, operational modes or activated navigational functions).

EP 4 The master should [be able to] [receive alerts and have] access [to] the alert management at all times. [The operator responsible of the MASS should be able to directly handle the alert management Human Machine Interface (HMI). It should be possible to provide the central alert management HMI at the control position for the operator.]

EP 5 Audible alerts should guide human operators to the task stations or displays which are directly assigned to the function generating the alert and presenting [at least] upon request the cause of the announcement and related information for decision support, e.g., dangerous target alarms should appear and have to be acknowledged at the workstation where the collision avoidance function is provided.

EP 6 As alerts can be displayed at several locations and task stations, the system should be consistent as far as practicable with respect to how alerts are [displayed] {presented}, silenced and acknowledged at any one task station. [Actions taken in a task station should be visible to all other relevant task stations.]

EP 7 [Means of] direct communication channels should be established and maintained between the person operating the MASS and any person on board [should be provided]

EP 7bis The applicability of conventional alerts should be reviewed in accordance with the autonomous level and operation mode of MASS. Specific considerations should also be given to the specific task/design of ROC.

Option 1 (IACS)

[EP 8 In addition to conventional alerts, specific consideration related to the operation of MASS, as outlined in Chapters of Part3, should include but not limited to :

- .1 upon entering a fallback state or upon recognizing the need to enter fallback state;
- .2 in case of equipment failure affecting operations of MASS functions or significantly increasing of risk of MASS operation e.g. loss of redundancy;
- .3 in case a system, that is critical to MASS operation, exceeds or is about to exceed its ODD. Some typical examples:
 - a. the ANS cannot make an appropriate collision avoidance plan;
 - b. the ANS cannot control the ship appropriately (e.g., deviation from the intended course and/or set speed range);
 - c. the ANS itself and/or any other systems connected to the ANS (including sensors, actuators, and communication systems) have any abnormalities or degradation;
 - d. any conditions are about to deviate, or have already deviated from the predefined operating conditions of the ANS;
 - e. the ANS detects undefined event (e.g., signal to which response is not defined);
 - f. the communication quality is found to be reduced to a level where ROC operators cannot perform their intended operations;
 - g. rolling accelerations or amplitudes exceed prescribed limits;
 - h. the alert management system is not working properly; and
 - i. a detected or suspected cybersecurity breach.]

Option 2 (existing)

EP 8 In addition to conventional alerts, alerts specifically related to the operation of MASS should be considered [including those required in Chapters of Part 3] such as:

- .1 upon entering a fallback state or upon recognizing the need to enter fallback state;
- .2 in case the ANS cannot make an appropriate collision avoidance plan;
- .3 in case the ANS cannot control the ship appropriately (e.g., deviation from the intended course and/or set speed range);
- .4 in case the ANS itself and/or any other systems connected to the ANS (including sensors, actuators, and communication systems) have any abnormalities or degradation;

- .5 in case any conditions are about to deviate, or have already deviated from the predefined operating conditions of the ANS;
- .6 in case the ANS detects undefined event (e.g., signal to which response is not defined);
- .7 in case the communication quality is found to be reduced to a level where ROC operators cannot perform their intended operations;
- .8 in case rolling accelerations or amplitudes exceed prescribed limits;
- .9 in case of equipment failure during mooring at the berth for ship operating and shore personnel;
- .10 in case of ROC-specific abnormal situations and conditions requiring attention (e.g. power failures, task station failures, communication system failures at ROC-level, software failures at ROC-level)
- .11 in case the alert management system is not working properly; and
- .12 in case a detected or suspected cybersecurity breach.

EP 9 Activated alerts should only be audible and visible to human operators operating the MASS emitting the alert.] [Alerts should only be presented for handling (e.g., acknowledgement or silencing) at task station(s) used by the operator in charge of the tasks related to the initiated alerts. Alerts may be presented visually for information at other tasks stations.] [Activated alerts should maintain operator focus by being audible and visible only to those directly operating or supervising the MASS .]

EP 10 When an emergency alarm is activated, a sufficient number of dedicated human operators including the master of the MASS should be operating the MASS until the emergency is over.

[EP 11 Abnormal situations and conditions concerning more than one MASS should be classified as alarms.

Proposal by the Secretariat: Suggested modifications to the text in document MSC 108/ WP7.

EP **12** In a situation where the MASS is not able to enter a fallback state when deviating from its operational envelope, an emergency alarm should be initiated.

EP **13** Task stations presenting alerts for multiple MASS should have the means to organize alerts per MASS and have the means to delegate alert handling for selected MASS to another task station.

EP **14** Information in fleet view applications must be organized such that logically connected information is presented together. Applications presenting alerts from multiple vessels should as default organize alerts per vessel. Relevant information relating to understanding the consequence of the alert should be logically grouped with the alert information or be available with a single operator action.]]

CHAPTER 15 HUMAN ELEMENT

15.1 General

15.1.1 This chapter provides principles and areas for consideration to ensure human element issues are effectively addressed, including the crew on board the MASS or operators in a Remote Operations Centre (ROC), as appropriate.

15.1.2 To effectively address human element issues, the operation of a MASS should consider the entire spectrum from human-machine collaboration/cooperation and expected/anticipated emerging interaction, to human activities performed by, and with effect on, shipowner, crew onboard a MASS, personnel in ROC(s), and other relevant parties.

[15.1.3 This chapter, as appropriate, may guide seafarers operating conventional ships and their shore-side personnel in comprehending the operation of a MASS.]

15.2 Roles and responsibilities

15.2.1 The allocation of tasks for personnel in relation to all MASS functions including roles and responsibilities should be defined according to the Concept of Operations (ConOps) and described in the task allocation summary defined in 1.7bis5.

15.2.2 Safe operation of a MASS is the responsibility of the designated Master regardless of the mode of operation, and they hold ultimate responsibility and authority over any operational decisions within a clear chain of command.

[15.2.3 The Master who is responsible for a MASS may be onboard the MASS or at a ROC.]

[15.2.4 If there is a crew or persons on board, a Master should be on board as well, to ensure the safety of personnel and operations.]

15.2.5 The Master should be appropriately supported by competent personnel to whom they may delegate specific tasks and responsibilities.

15.2.6 Several Masters could be operationally responsible for a MASS on a single voyage, under certain conditions, while only one Master should be responsible at any given time.

[15.2.7 Crew onboard should have overriding authority over any remote operators with respect to the safety of persons onboard and operations.]

15.2.8 MASS remote and onboard crew should have access to inputs providing sufficient situational awareness according to the ConOps of the MASS and general intervention capabilities over functions and the proper means to perform such intervention.

15.2.9 To ensure the safety and security of crew or any other persons onboard, a clear contingency/emergency plan should be in place and an onboard responsible person should be designated along with clearly defined responsibilities and authority.

15.3 Supplementary Competencies, Training and Familiarization

15.3.1 The Master and all personnel involved in MASS operations should complete appropriate training and be suitably experienced, with focus on where the roles and responsibilities differ from conventional shipping.

15.3.2 Administrations should ensure that all personnel involved in MASS operations are certified according to the STCW Convention and the STCW Code, as amended, with supplementary competencies as necessary relevant to the operation of MASS.

15.3.3 Administrations should ensure, all personnel employed or engaged in the oversight, monitoring, control or operation of the functions of a MASS should also have undertaken system-specific training.

15.3.4 A Remote Operator(RO) of a MASS should have competencies and experience as an officer qualified in accordance with appropriate STCW requirements.

15.3.5 The shipowner should establish procedures to ensure that new personnel and personnel transferred to new assignments related to the ConOps of the MASS are given proper familiarization with their duties. Instructions which are essential to be provided prior to undertaking operations should be identified, documented and given.

15.3.6 Administrations should consider clearly evidenced remote operational experience as sea time equivalency for those working within a ROC for the revalidation of [Certificate of Competency's / MASS Operator Endorsement] where appropriate.

15.3.7 Supplementary competencies, training and familiarization provided should be adequate for all personnel to fully demonstrate their roles and responsibilities, including but not limited to:

- .1 knowledge and competencies to operate remotely operated or autonomous system(s) onboard MASS and within a ROC(s), as appropriate;
- .2 role allocation between remotely operated or autonomous systems and seafarers onboard a MASS and personnel at ROC(s);
- .3 knowledge and familiarization of the Operational Design Domain (ODD) of the remotely operated or autonomous systems of the MASS;
- .4 ability to determine and apply an appropriate level of hazard identification and risk assessment before performing any operation of the MASS;
- .5 ability to takeover manual control of a MASS when remotely operated or autonomous systems malfunction and/or depart from the ODD (reference to Part 2, chapter 3 and chapter 4.6);
- .6 knowledge and skills to safely work with MASS machinery, automation technology, electrical and ICT systems, including Human-Machine Teaming, and interaction with any shore-side operations;
- .7 an ability to conduct clear and effective communications with other ROCs, ships and relevant stakeholders, such as Vessel Traffic Service, Pilotage, Port Authorities; and
- .8 an ability to perform functions assigned to them, irrespective of the location of a control station.

Alternative proposal

Proposal by the Secretariat: Suggested modifications to the text in document MSC 108/ WP7.

15.3 Familiarization and supplementary competencies

15.3.1 Introducing advanced technological tools in shipping without providing sufficient training to the crew and staff can lead to underutilization or misuse of the technology. This can negate the potential benefits of the technology and even increase the risk of accidents or inefficiencies.]

15.3.2 [Noting the Principles of the Code [(part 1, 2 Principle)] ensure the base IMO instruments are maintained,] this section provides principles for [new][supplementary] competencies, and corresponding training and familiarization provisions required when operating a MASS to fully demonstrate the roles and responsibilities including role allocation between the autonomous or automated system and for seafarers and operators on both the MASS and ROC(s).]

15.3.3 The Remote Operator of a MASS should have competency and experience as an officer qualified in accordance with appropriate STCW requirements. [The Administration may allow the Remote Operator who has equivalent competency and experience to the satisfaction of the Administration.]

Familiarization for seafarers on board MASS

15.3.4 Seafarers on board MASS should be familiar with the operation of the autonomous or automated systems, for which the seafarers are responsible, installed on the MASS.

15.3.5 Seafarers on board MASS should understand the Operational Design Domain (ODD) of the MASS systems properly.

15.3.6 Seafarers should immediately take over the control of the MASS from the system and should command manually when the autonomous or automated systems malfunction and have departed from the ODD.

Familiarization for Remote Operator of a MASS

15.3.7 The Remote Operator should be familiar with the operation of the autonomous or automated systems, for which the seafarers are responsible, installed on the MASS.

15.3.8 The Remote Operator should be familiar with the ODD of the MASS systems properly.

15.3.9 The Remote Operator should be familiar with the immediate takeover of control of the MASS from the system and should be familiar with a command manually when the autonomous or automated systems malfunction and depart from the ODD.

Supplementary competencies for seafarers on board a MASS

15.3.10 Seafarers on board a MASS should be competent in areas including, but not limited to:

MSC 108/4/13 Annex, page 2

- .1 knowledge of the autonomous or automated systems on board MASS, in association with situational awareness, collision avoidance, and auto-tracking of the intended route, cybersecurity, propulsion plant etc.;
- .2 demonstration of the roles and responsibilities of seafarers on board MASS, including role allocation between the autonomous or automated system and seafarers on board;
- .3 knowledge of the ODD of the autonomous or automated systems of the MASS;
- .4 ability to apply the risk assessment on any operation of the MASS; and
- .5 managerial skills against the considerable risks while using the autonomous or automated systems. (Especially, the dangers of over-reliance on automated systems).

Supplementary competencies for the Remote Operator of MASS

15.3.11 Remote Operators should be competent in areas including, but not limited to:

- .1 knowledge of the autonomous or automated systems on board MASS and of RCC in association with situational awareness, collision avoidance, and auto tracking of the intended route, cybersecurity, propulsion plant etc.;

- .2 demonstration of the roles and responsibilities of RCO including role allocation between the autonomous or automated system and seafarers on board;
- .3 knowledge of the ODD of the autonomous or automated systems of the MASS
- .4 ability to apply the risk assessment on any operation of the MASS; and
- .5 managerial skills against the considerable risks while using the autonomous or automated systems (especially, the dangers of over-reliance on automated systems) .

15.3.12 Administrations should consider clearly evidenced remote operational experience as sea time equivalency for those working within a ROC for the revalidation of [Certificate of Competency/MASS Operator Endorsement] where appropriate.]

15.4 Manning/crewing

15.4.1 When determining the operationally safe manning levels onboard a MASS, the shipowner should take into account the following:

- .1 as stated in IMO instruments as well as any relevant international instruments concerning manning levels*; and

*footnote: SOLAS Chapter IX, ISM Code, SOLAS Chapter V, A.1047(27), STCW, Res.A.1047(27), MLC 2006, as amended

- .2 as described in the ConOps and the hazard identification and risk assessment, sufficient to respond to emergencies, and taking into account the task allocation summary.

15.4.2 When determining the operationally safe manning levels in a ROC or a control station, shipowner should take into account the following:

- .1 the Safe Manning Document that details the agreed minimum levels of safe manning for each MASS that the ROC is expected to operate;
- .2 optimal working conditions, configurations and ergonomics for all Human-Machine interfaces for the expected tasks to be performed by the human operator;
- .3 the relative levels and structure of safe operational manning between shipboard and shore-side personnel;
- .4 the operational emergency response plan for each MASS that the ROC is operating (reference to Part 2, Chapter16); and

15.5 Occupational safety and health

15.5.1 To ensure the health, safety, and comfort of any personnel on board a MASS or at a ROC, the followings should be considered:

- .1 appropriate medical care or aid, sanitary conveniences and suitable facilities for well-being of personnel;
- .2 sufficient and unobstructed space to allow free movement for personnel;

- .3 need to specify a person responsible for the implementation of, and compliance with, the occupational safety and health policy for the MASS operation, considering both shipboard and shore-side personnel;
- .4 safe hours of work, hours of rest and breaks, including periods of natural light and fresh air throughout the duration of watches to ensure sufficient rest and to limit fatigue for ROs that are undertaking watchkeeping duties; and
- .5 a cultural atmosphere regarding safety that encompasses responsible behaviour and trust whereby persons are encouraged to provide essential safety-related information without fear of retribution.

15.5.2 Administration should apply appropriate occupational health protection policy of the territory in which the ROC is situated.

Proposal by the Secretariat: Suggested modifications to the text in document MSC 108/ WP7. To be further considered whether to keep 15.6.1 numbering as it is not an FR but rather an introductory text.

15.6 Application of Human Centred Design (HCD)

15.6.1 This section provides the perspectives for the interaction between a system and an operator. The interface is the aggregate of means by which the users interact with a machine, device, and system (the System). The interface provides means for input, allowing the users to control the system and output, allowing the system to inform the users.

15.6.2 Human-System Interaction should be robust. This should include intervention strategies for reasonably foreseeable situations (normal, abnormal and emergency) and reversionary means of control based on the principle of what the System can provide to humans. The design processes should consider resilience, reliability, trust and sustainability of the whole Human-Machine System including the use of new innovations.

15.6.3 The design and operations of the MASS and associated ROC human-machine interfaces should be focused on assistance to the human operators. Human operators should be involved in validation stages to ensure reliability and safety.

15.6.4 The design and layout of the HMI of a MASS or a ROC, and the testing of the working human-machine system, should be based on the principles in Part 2, chapter 3, taking into account the relevant instruments* adopted by the Organization.

*footnote: SOLAS regulation V/15, SN.1/Circ 265 and MSC.1/Circ.1512

15.6.5 Based on an assessment of user tasks and the configuration of systems, and taking into account all HMIs, full consideration should be given to achieving the optimal working location and conditions used for the control or monitoring of a MASS operation.]

CHAPTER 16 MAINTENANCE AND REPAIR

16.1 Goal

The goal of this chapter is to provide for maintenance and repair to ensure that all [HME systems] [structure, machinery, and equipment] will remain fit to proceed to sea, without danger to the ship or persons on board,

16.2 Functional requirements

To achieve the above-mentioned goal the following functional requirements are embodied in this chapter.

FR 16.2.1 Maintenance of Computer-based integrated systems should ensure safe operations in normal and emergency situations.

FR 16.2.2 Suitable monitoring and control capability should be provided to ensure system and machinery faults are detected in normal and emergency conditions and can maintain any [fallback states].

FR 16.2.3 Suitable redundancy actions should be provided taking into account the number of qualified persons on board that are available to respond to system and machinery faults.

FR 16.2.4 Maintenance requirements for the equipment and systems used on board should not be compromised by ships mode of operation.

FR 16.2.5 Qualified [authorized] persons should be available to remotely monitor system and equipment faults and abnormal conditions to verify their cause and confirm that the designed redundancy has been effective in maintaining the intended performance.

PART 3 GOALS, FUNCTIONAL REQUIREMENTS AND EXPECTED PERFORMANCE

GENERAL

Each chapter in this part consists of the Goal of the chapter, Functional Requirements (FR) to fulfil the Goal, and the Expected Performance (EP) associated with those functional requirements.

Chapters should be applied in full, but application of specific chapters may be waived in agreement with the Administration as part of the approval process depending on the ConOps and/or MASS functions implemented.

A ship should be considered to meet a functional requirement set out in this part when either:

- a) the ship's design and arrangements meet all the expected performances associated with that functional requirement; or
- b) part(s) or all of the ship's relevant design and arrangements have been reviewed and confirmed to be in accordance with regulation [X] of SOLAS chapter [Y], and any remaining parts of the ship meet the relevant expected performance.

The functional requirements and associated expected performances should take into account the modes of operation of the ship and depending on human presence on board.

CHAPTER 17 SAFETY OF NAVIGATION

17.1 Goal

The goal of this chapter is to provide for safe navigation.

17.2 Functional requirements

To achieve the above-mentioned goal, a ship should comply with all relevant requirements for safety of navigation in SOLAS, as modified and/or supplemented by the functional requirements of this chapter, and COLREG.

17.3 General

A MASS should comply with the following functional requirements for navigation in general.

17.3.1 The navigation equipment and systems for MASS operations should be designed, constructed, and installed to maintain their functionality under the expected conditions in the OE.

- EP 1 An ANS or system for remote navigation should be installed enabling the functionalities described in the ConOps, other design documents and the task allocation summary.
- EP 2 The use of remotely controlled or autonomous systems for non-navigation functions or tasks should not endanger the safe operation of an ANS or system for remote navigation.
- EP 3 An ANS or system for remote navigation should effectively integrate and coordinate with other ship systems to prevent conflicting operational commands.

- EP 4 In the event of failure of an ANS or system for remote navigation, the ship should be able to be operated safely by alternative means.
- 17.3.2 All data necessary for safe navigation should be available, in an appropriate format.
- EP 1 A MASS should meet the requirements of SOLAS regulations V/19.2.1.4 and 19.2.1.5 by electronic means.
- EP 2 Data used by or for input into an ANS or system for remote navigation should be in a machine-readable format.
- EP 3 The voyage plan should be accessible, at all locations where navigation tasks are executed

17.4 Sub-functions for MASS navigation

Any or all of the navigation sub-functions below could be autonomous or remotely controlled, while having different modes of operation according to the ConOps:

NOTE: The terms “automated/autonomous” and “remotely controlled/operated” should be finalised

- .1 Voyage planning: FR17.4.1-.2
- .2 Situational awareness: FR17.4.3;
- .3 Collision and grounding risk avoidance: FR17.4.4; and
- .4 Route execution and monitoring: FR17.4.5

As such, a ship should comply with the relevant sub-function related requirements only when that sub-function is planned to be autonomous or remotely controlled.

Voyage planning

17.4.1 An ANS or system for remote navigation that is provided with the voyage plan, should have a means to ensure that the voyage plan is safe for navigation.

- EP 1 An ANS or system for remote navigation should provide a means to review and as necessary modify the voyage plan at all times.
- EP 2 An ANS or system for remote navigation should be capable of confirming to the Master that the voyage plan has been correctly received and is able to validate and implement the voyage plan received.
- EP 3 The use of the voyage plan, and any modification of the voyage plan, by the ANS or system for remote navigation should not be possible without an approval, including verification of its correctness, by the Master.

17.4.2 An ANS or system for remote navigation that is capable of developing the voyage plan should have a means to ensure that the voyage plan is safe for navigation, taking into account the Guidance developed by the Organization*.

* Resolution. A.893(21)

- EP 1 An ANS or system for remote navigation should be provided with access to all nautical and hydrographic data necessary to appraise and plan a voyage.
- EP 2 The voyage plan developed by an ANS or system for remote navigation should be presented in a form that allows the Master to approve the plan.

- EP 3 The voyage plan should provide an ANS and/or system for remote navigation and Master with all information necessary to execute and monitor a voyage.
- EP 4 An ANS or system for remote navigation should provide a means to review and as necessary modify the voyage plan at all times.

Situational awareness

17.4.3 An ANS or system for remote navigation should maintain adequate situational awareness for the purpose of ensuring safe navigation.

- EP 1 An ANS or system for remote navigation should continuously monitor all information necessary for safe navigation, based on the operational envelope.
- EP 2 An ANS or system for remote navigation should continuously analyze the current situation in order to perceive and process navigational and environmental elements and project or predict future statuses. This should be done whilst considering the operation goals and objectives, the system's capability, and the mode of operation.
- EP 3 An ANS or system for remote navigation should provide a means to make the information obtained and all analysis accessible as necessary.

Collision and grounding avoidance

17.4.4 An ANS or system for remote navigation should ensure a means to take action in accordance with COLREG to prevent collisions, as well as allisions and groundings.

- EP 1 Planning and decisions to alter course and/or speed should be both timely and in accordance with safe operating limits of ship.
- EP 2 Any action to avoid collisions, allisions or groundings should be based on an assessment of the risk and the action necessary to avoid the risk of a collision, allision or grounding.
- EP 3 Action to avoid collisions, allisions or groundings should not result in another situation which presents intolerable risk to the ship, other ships or the marine environment.

Route execution and monitoring

17.4.5 An ANS or system for remote navigation should safely execute the voyage plan and appropriate monitoring should be provided.

- EP 1 An ANS or system for remote navigation should execute the planned route within the pre-defined limits of allowable deviation taking into account the factors listed in the Guidance developed by the Organisation*.

* Resolution. A.893(21)

- EP 2 An ANS or system for remote navigation should notify the Master if at any time the prevailing circumstances and conditions necessitate a deviation from the voyage plan.
- EP 3 An ANS or system for remote navigation should continuously monitor the progress of the ship against the voyage plan and make the information available, as necessary.

NOTE: Following paragraph might be moved to other chapter of part 2

[Chapter 2] Proper records relating to navigation should be stored appropriately in order to contribute to safety of navigation and casualty investigations.

EP 1 Records of the movements, activities and time relating to an ANS [or system for remote navigation] should be maintained at the same level as that in voyage data recorders.

[EP 2 [In the case of MASS without crew on board,] records of navigational activities and daily reports should be automatically stored on board and at the ROC as appropriate.]

[EP3 Operation in a degraded state or executing a fallback response, and time of those events, relating to an ANS [or system for remote navigation] should be automatically stored on board and at the ROC, as appropriate.]

17.5 Override

An ANS or system for remote navigation should be capable of being overridden at all times from location(s) where control of a ship's navigation can be exercised.

EP 1 Means for overriding operation of an ANS or system for remote navigation should be simple to operate, independent of the systems that they control and allow for control to be taken immediately.

CHAPTER 18 REMOTE OPERATIONS

18.1 Goal

<option 1>

The goal of this chapter is to ensure the safe remote operation of MASS [, or automated functions thereof,] from a location which is not on board the ship, taking into account the modes of operation, the number of persons on board, and the total number of MASS that are operated from the same location.

<option 2>

[The goal of this chapter is to ensure the safe, secure and effective remote operation of MASS [, or automated functions thereof,] when duties and responsibilities for safe operation are assigned to a Remote Operations Centre (ROC),] [taking into account the modes of operation, the number of persons on board, and the total number of MASS that are operated from the same location.]

18.2 Functional Requirements

To achieve the above-mentioned goal, a ship and ROC should meet the functional requirements of this chapter.

18.2.1 A [location/ROC] should be provided to ensure safe, secure, and effective MASS operations [or the automated functions thereof] at any time.

EP 1 A [location/ROC] should have:

- .1 facilities that are secure and protected from unauthorized access.

- .2 means to enable reliable connectivity and communication between ROC(s) and the ship, third parties and persons on board.
- .3 facilities to allow access to, and sharing of, certificates and other documents required to demonstrate that the ship is compliant with international, national and regional requirements.
- .4 arrangements, such that the failure and [subsequent] recovery of the ROC would not result in an unsafe state or intolerable risk on or around the ship in service [, including the use of redundancy or enter a fallback state].
- .5 validated and verified systems to support the effective operation of the ship.
- .6 sufficient and relevant qualified personnel [in accordance with Management of Safe Operations requirements] to enable safe MASS operations.
- .7 facilities to ensure data and information used, produced, sent or received is retained in reliable and tamper-proof storage and at a suitable standard of data quality, and referring to the SOLAS requirements for Voyage Data Recorders.

18.2.2 A control station(s) should be provided to ensure the safe, secure and effective MASS operations [or the automated functions thereof].

EP 1 A control station should:

- .1 have appropriate validated and verified systems to enable effective operation of MASS.
- .2 provide sufficient and accurate data and information to enable the remote operator(s) to carry out their role(s) effectively.
- .3 be fully compatible with the ship throughout its operational life [or the automated functions] under its control.
- .4 be tested to ensure that when installing and updating system(s) that the related onboard equipment and devices have appropriate compatibility and interoperability.
- .5 ensure failure and recovery of the control station(s) would not result in an unsafe state or intolerable risk, on or around the ship, including the use of redundancy and back up measures. [OR have mechanisms by which the control station can enter a fallback]
- .6 be designed and operated in such a way that its location does not result in loss of control or negatively affect the ship.

18.2.3 Validated and verified systems and interfaces between control station(s) and the ship should be provided to ensure the remote operator can operate the ship safely, securely and effectively.

EP 1 This will be accomplished by ensuring the remote operator is able to:

- .1 keep a watch at sea or in-port in a manner conforming to the principles of watchkeeping;

- .2 send and receive sufficient and accurate information/commands effectively and securely between the ROC, the ship, third parties, and any shipboard personnel;
- .3 take all decisions necessary to ensure the safe operation of the ship.
- .4 know the status of the connectivity at the control station(s) and the ship and, where relevant, by third parties.
- .5 know which systems can be controlled, and have clear visibility of the current control position [OR have the location which is in control clearly visible].
- .6 know when conditions on the ship or at the ROC deviate from the OE.
- .7 monitor the condition and mode of operation of ship equipment and systems and take measures to prevent and/or rectify deficiencies when [alerted][warnings are actuated].

18.2.4 The transfer of operation of the ship [, or the automated functions thereof] should be safe and secure to ensure safe navigation.

EP 1 Safe transfer of operation of the ship will be accomplished by ensuring:

- .1 transfer and synchronization of all necessary information is possible between control station(s), ROC and the ship;
- .2 operation can be transferred safely and securely during failure and/or recovery or an emergency situation at the ROC or control station(s);
- .3 the control is not [provided][possible] by multiple positions at the same time and the present control station is clearly indicated both in ROC and on board the ship;
- .4 there is no loss of control of the ship when the operation is transferred.

18.2.5 Software used in the control station(s), ROC and/or on board the ship should be appropriately managed and remain within the defined OE to ensure safe, secure and effective operation.

EP 1 The software used should be:

- .1 designed, integrated, managed, maintained and supported throughout its operational life;
- .2 able to receive, recognize and assist with the prioritization of emergency and non-emergency situations, occurring on board the ship to enable the remote operator(s) to carry out their role(s) effectively; and
- .3 designed to ensure that the remote operator(s) are able to read and understand the information transmitted to the ROC.

CHAPTER 19 STRUCTURE, SUBDIVISION, STABILITY AND WATERTIGHT INTEGRITY

19.1 Goal

The goal of this chapter is to ensure that the strength, subdivision, stability, and watertight integrity are within acceptable limits.

19.2 Functional Requirements

To achieve the above-mentioned goal, a ship should comply with all relevant SOLAS structural, stability, subdivision and watertight integrity requirements, as well as the 1966 International Load Line Convention/1988 Protocol and the 2008 Intact Stability Code, as modified and/or supplemented by the functional requirements of this chapter.

[19.2.1 Any automated/autonomous function for the control of intact and damaged stability should be provided with an independent automated/autonomous supervising function. The action of the independent supervising function should be triggered by failures/events (*) of the stability control system.

19.2.2 The independent supervising function should be provided to detect and respond to subdivision, stability and watertight integrity issues and failures and should be able to monitor, control, and operate systems on board that may affect the stability of the ship.]

19.2.2 The automated/autonomous functions, supervising functions, and functions carried out at the ROC should provide means to ensure compliance with the relevant intact stability criteria as set out in the list of all relevant applicable stability regulations of chapter 3.4.

EP 1 The automated/autonomous functions, supervising functions, and functions carried out at the ROC should be:

- .1 capable of continuously determining the ship's intact stability and assessing survivability during operation;
- .2 capable of continuously control the loading conditions, the longitudinal and local strength;
- .3 capable to assess the survivability of the ship in case of damage, to maintain the ship, at all times is operating within the stability envelope as prescribed in the stability booklet;
- .4 capable of restoring compliance with relevant intact stability requirements if the system has detected that these requirements are not met;
- .5 able to bring the ship to the appropriate fallback state upon activation of a critical alarm related to the intact stability; and
- .6 able to detect existing or predictable stability failures and raise alarms.

19.2.3 The supervising function, as the dedicated monitoring function, is explicitly responsible or continuously monitoring system performance, detecting failures/events of the automated/autonomous stability function, as well as other failures/events, and promptly initiating corrective responses. It should achieve this through active oversight, relying on an independent measuring system, provided with a comprehensive array of sensors.

19.2.4 The stability control system and the supervising function should be resilient to single failure.

19.2.5 The independent measuring system and sensors should be designed and maintained to consistently provide accurate and reliable data to ensure that the ship stability, subdivision, and watertight integrity is maintained under all conditions.

19.2.6 The independent measuring system and sensors should incorporate redundancy mechanisms to ensure data availability in the event of component failure.

- EP 1 Regular calibration and maintenance procedures should be in place to ensure accuracy over time.
- EP 2 Data from the measuring system and sensors should integrate seamlessly with the supervising function and other systems.
- EP 3 The system should alert operators in case of issues with the measuring system or sensors.
- EP 4 The measuring system and sensors should be compatible with the ship's overall stability and safety systems.
- EP 5 Comprehensive documentation should be available detailing specifications, maintenance procedures, and testing protocols.

19.2.7 The stability control system should provide measures to prevent excessive motions of the ship in adverse sea conditions.

- EP 1 To prevent excessive motions, the stability control system should be able to:
 - .1 continuously measure amplitudes and accelerations of ship motions;
 - .2 detect when the ship's motions exceed predefined thresholds, triggering alarms on board and/or at the ROC, while recording readings from the sensors;
 - .3 adjust speed and course from the ship and/or the ROC, in response to the wave direction to ensure compliance with predefined limits for the ship's movements; and
 - .4 predict the combination of ship's speed and course, under current weather conditions, that may cause the ship to exceed the limits, and initiate appropriate corrective mitigation actions from the ship and/or at the ROC.

CHAPTER 20 FIRE PROTECTION, FIRE DETECTION AND FIRE EXTINCTION

20.1 Goal

The goal of this chapter is to ensure that fire-safety systems and equipment are effective.

20.2 Functional requirements

To achieve the above-mentioned goal, a ship should comply with all relevant SOLAS fire safety objectives and requirements as modified and/or supplemented by the functional requirements of this chapter.

20.2.1 A ship should remain under control or enter a fallback state during and following a fire event in any single compartment that is directly linked to the control of the ship.

20.2.2 A fire which is limited to a single compartment that does not have a direct link to the control of the ship, should not cause a loss of control or lead to a fallback state.

20.2.3 Means should be provided to detect, confirm, and locate a fire incident.

- EP 1 All alarms related to fire safety systems should be routed to control station(s).
- EP 2 Means for timely detection of a fire should be provided in all spaces with a fire risk.
- EP 3 An operator should be made aware of the detection and location of a fire along with the status of, and any actions taken by, fire protection systems.
- EP 4 After a fire detection alarm is activated, means should be provided to confirm a fire using a different detection source.
- EP 5 If alarm signals are not acknowledged, a secondary alarm should be automatically activated at control station(s) and throughout the ship.
- EP 6 After detection and confirmation of the fire, means should be provided to locate the seat of fire, so that the most appropriate fire-extinguishing means may be activated.

20.2.4 Means should be provided to enable the appropriate use of fire-extinguishing systems, taking into account possible human presence.

- EP 1 Information and instructions in relation to fire safety should be provided to any personnel boarding the ship.
- EP 2 Fire extinguishing systems should be able to be safely isolated and verified prior to compartment access.
- EP 3 Fire extinguishing systems should provide onboard indication and warning prior to, and during, activation.
- EP 4 The stability of the ship should be monitored during any fire-fighting operation.

NOTE: When discussing chapter 19, appropriate location of EP 4 should be considered.

20.2.5 Means should be provided to monitor and assess fire growth and fire-fighting effectiveness during and after fire.

- EP 1 Means should be provided to assess any smoke development.
- EP 2 Means should be provided to assess the temperature development in spaces adjacent to the compartment affected by the fire.

20.2.6 Means should be provided to enable the remote and onboard control of all active fire protection measures, including containment measures.

- EP 1 All active fire protection measures should be individually controllable, allowing activation, deactivation, and status monitoring at control station(s).
- EP 2 Means should be available to detect fire protection systems related faults and provide indication at control station(s).

20.2.7 Means should be provided to facilitate an intervention from external fire responders.

- EP 1 Procedures should be in place to transmit any relevant information and data to external fire responders during and following a fire incident.
- EP 2 Means of communication between the external fire responders and the ship and ROC should be ensured during and following a fire incident.

CHAPTER 21 LIFE-SAVING APPLIANCES AND ARRANGEMENTS

21.1 Goal

The goal of this chapter is to provide for safe escape, evacuation, [and survival].

21.2 Functional requirements

To achieve the above-mentioned goal, a ship should comply with the life-saving appliances and arrangements requirements in SOLAS as modified and/or supplemented by the functional requirements of this chapter.

21.2.1 All personnel involved in the operation of the ship should be trained to take appropriate measures in case persons on board are required to abandon the ship.

21.2.2 The use of automated and/or remotely controlled life-saving appliances should not endanger the safety of any persons.

21.2.3 In addition to the ship, ROC personnel involved in the operation of the ship should be provided with instructions, information and training in the operation of the ship in relation to life-saving appliances and their use.

21.2.4 As appropriate, survival craft and life-saving appliances, and necessary equipment should have the capability to operate automatically to enable the safe abandonment of persons from the ship.

[21.2.5 Provision should be made to enable the deployment of life-saving appliances and response to be undertaken on board by an external responder. This should include provisions for establishing communications with the ROC and a response to an abandonment of personnel.]

21.2.6 Autonomous or remotely controlled life-saving appliances should not present an increased risk to the safety of persons on board.

21.2.7 An ROC should be capable of taking measures to facilitate abandonment of the ship and be provided with sufficient information to assess the risks to personnel in order to manage appropriate response actions.

21.2.8 An appropriate level of communication between the ship and the ROC should be maintained during and following an abandonment of the ship.

CHAPTER 22 SPECIAL MEASURES TO ENHANCE MARITIME SECURITY

22.1 Goal

The goal of this chapter is to ensure adequate security.

22.2 Functional Requirements

To achieve the above-mentioned goal, the ship should comply with the requirements of the special measures to enhance maritime security in SOLAS and the ISPS Code as modified and/or supplemented by the functional requirements of this chapter.

22.2.1 Means should be provided to enable the assessment of security effectiveness for autonomous and remotely controlled systems.

22.2.2 Onboard and remote management of autonomous systems should be provided to enable control of the systems.

22.2.3 The ship should remain under control during and following a security event.

22.2.4 The ship should be able to communicate and exchange security-related information with the ROC, taking into account the sensitivity of the information and authorization to access security-sensitive information.

22.2.5 The use of autonomous security systems should not prevent the effective physical security; structural integrity; personnel protection systems; procedural policies; radio and telecommunication systems including computer systems and networks; and other areas that may, if damaged or used for illicit observation, pose a risk to persons, property, or operations on board the ship.

22.2.6 There should be a mechanism for safely shutting communications down when the security of the ROC has been compromised.

22.2.7 Upon identification of a security event the ship should enter an appropriate fallback state and be capable of maintaining that state during and following the event to the degree necessary.

22.2.8 An appropriate level of communication between the ship and the ROC should be maintained during and following a security event.

CHAPTER 23 SEARCH AND RESCUE

23.1 Goal

The goal of this chapter is to ensure that the duties and tasks regarding Search and Rescue are fulfilled.

23.2 Functional requirements

In order to achieve the goal set out in paragraph 9.1 above, the following functional requirements are embodied in the provisions of this chapter:

23.2.1 Every ship which is in a position to be able to provide assistance on receiving information from any source that persons are in distress at sea, should proceed with all speed to their assistance, if possible informing them or the search and rescue service that the ship is doing so.

- EP 1 The ship should be able to receive distress information from any source, included search and rescue services.
- EP 2 In order to know the ship's ability to render assistance, means should be provided to correlate the ship's own status, and position to any distress signal.
- EP 3 If its cooperation is required, the ship should be able to coordinate with relevant rescue coordination centres.
- [EP 4 The ship should be able to effectively use the rescue devices applicable to its type of ship, independent of whether there is crew on board.
- EP 5 Remotely or automatically operated rescue devices should not expose persons in distress to a greater danger.
- EP 6 The ship should have a space on board sheltered from harsh meteorological conditions to accommodate retrieved persons in distress until it is able to deliver them to a safe place and the possibility of communication with ROC should be assured.

- EP 7 Means should be available to prevent the access of rescued persons to unauthorized spaces.]

23.2.2 Every ship should be able to detect and locate distress signals, including audio and visual alerts as per SOLAS distress signals list.

- EP 1 The ship should be able to identify distress signals of COLREG Annex IV.
- EP 2 The ship should be able to locate distress signals with bearing and distance or with latitude and longitude.
- EP 3 The ship should be able to navigate to the point where distress signal is located.
- EP 4 Environmental and internal factors that may affect the ship's speed should be taken into account.
- EP 5 The ship should be able to detect distress communications, directly and through an ROC.

[23.2.3 In case of distress, if required, the master should be able, even remotely, to lead SAR on-scene commander duties.

- EP 1 Means to ensure that the master is able to lead SAR activities, including communication and [local] coordination of surface search, should be provided.
- EP 2 Environmental and other factors that may affect the ship's capability to lead on-scene activities should be addressed.
- EP 3 Master's authority and responsibility to make decisions on SAR operations, should be assured.
- EP 4 IAMSAR Manual Volume III (Mobile Facilities) should be available to the Master, even if controlling the ship remotely.

23.2.4 After a collision with other ship, every MASS should render assistance and provide information.

- EP 1 Means to address the capability of ship to render assistance to the other ship, its crew and its passengers after a collision should be provided, included damage sustained and environmental factors.
- EP 2 Ship should be able to identify the assistance that's able to render, according to internal and environmental data.
- EP 3 Means to provide the name of the vessel, its port of registry and the next port of call to other vessel should provided.

23.2.5 In case of distress, if required, a MASS Master should be able to lead on-scene SAR activities.

- EP 1 Means to ensure that the Master is able to lead SAR activities, including communication and coordination of surface search, should be provided.
- EP 2 Environmental and internal factors that may affect the ship capability to lead on-scene activities should be addressed.
- EP 3 Master's authority and responsibility to make decisions on SAR operations, should be assured.

- EP 4 IAMSAR Manual Volume III (Mobile Facilities) should be available to the Master, even if controlling the ship remotely.”

23.2.6 MASS, with or without crew on board, should be able to maintain distress communications (directly and through ROC).

- EP 1 MASS should be able to emit distress communications, including ship to shore distress alerts (according to Communications chapter 3).
- EP 2 MASS should be able to keep an operation watch at the distress frequencies.
- EP 3 MASS should be able to immediately identify distress communications.
- EP 4 Depending on the distress message, means to promptly identify the type and scale of the emergency should be provided.
- EP 5 MASS should be able to receive distress communications.
- EP 6 MASS should be able to relay distress communications.
- EP 7 MASS should be able to silent distress communications as applicable.
- EP 8 Means to report SAR information to MRCC should be available.
- EP 9 Means to ensure that alerts are transmitted when required and that false alerts are avoided should be available.

23.2.7 A MASS, with or without crew on board, should have means to recover persons in distress.

- EP 1 Means to recover persons could be operated even if there are no persons on board.
- EP 2 Means to recover should adapt to environmental conditions.
- EP 3 Means to recover persons should ensure easy boarding.
- EP 4 Means to recover persons should not expose persons in distress to a greater danger than that is intended to avoid.
- EP 5 MASS should have, included in the emergency management system, specific plans, procedures and training and drills for the rescue of persons in distress, as well as manuals available to the master and officers in charge of the MASS.
- EP 6 MASS should have a maintenance plan for the means of retrieval from the water of persons in distress.
- EP 7 MASS should have a sheltered space on board from harsh meteorological conditions to accommodate retrieved persons in distress until is able to deliver them to a place of safety.
- EP 8 Means to prevent the access of rescued persons to unauthorized spaces should be available (according to Security chapter 9.3 Specific Requirements: Implementation of ship security plan).

23.2.8 A MASS should have specific plans and procedures, including responsibilities, for its own distress situations in the emergency management system.

- EP 1 MASS should be able to identify an internal distress situation according to its own internal and environmental sensors.

EP 2 Procedures, training and drills to address internal distress situations should be available.]

23.2.9 Every MASS vessel with persons on board should have a responsible master on board for leading activities on board MASS during distress situations. Master of the ship should also facilitate safe access and guidance of external rescue staff on board.

CHAPTER 24 CARGO HANDLING

24.1 Goal

The goal of this chapter is to provide for the handling and care of cargoes in a manner that ensures that the ship, persons on board and the environment are safe.

24.2 Functional requirements

To achieve the above-mentioned goal, the ship should comply with the functional requirements of this chapter.

24.2.1 Where the ship is either autonomous or remotely operated, consideration should be given to the need for any special arrangements regarding:

- .1 transfer of cargo information;
- .2 cargo handling;
- .3 cargo monitoring; and
- .4 emergency response for special categories of cargoes.

CHAPTER XX PERSONNEL SAFETY AND COMFORT

N.B. This chapter is being absorbed, as appropriate, by chapter 2.7 (Human element)

CHAPTER 25 TOWING AND MOORING

25.1 Goal

The goal of this chapter is to ensure safe and secure towing and mooring operations.

25.2 Functional Requirements

To achieve the above-mentioned goal, the ship should comply with the functional requirements of this chapter.

25.2.1 Shipboard mooring and towing arrangements should enable the ship to conduct berthing, un-berthing and towing functions [in all reasonably foreseeable conditions and circumstances][, taking into account the fallback states].

25.2.2 Means should be provided for effective coordination and conduct of mooring and towing operations.

25.2.3 Means should be provided to ensure the continuous monitoring/control capability for towing and mooring arrangements [during normal conditions][in all reasonably foreseeable conditions and circumstances] [, taking into account the fallback states].

25.2.4 Means should be provided to ensure that sufficient information about mooring and towing arrangements at marine facilities, terminals, and berths is available to enable the operations to be planned and conducted with due consideration to safety of property and personnel, and as appropriate, environmental protection.

25.2.5 Means should be provided to maintain safe berthing, un-berthing and towing, including its control/monitoring system, in the event of failure of the relevant equipment and system.

- [EP 1 If the mooring and towing equipment deviates from its ODD, the [operator] should be notified and appropriate action should be carried out to maintain the [operability of the ship].
- EP 2 If the mooring and towing functions fail in a way to force the ship to deviate from its OE, a fallback response should be carried out.]

CHAPTER 26 MACHINERY INSTALLATIONS

26.1 Goal

The goal of this chapter is to provide for safe and reliable machinery installations.

26.2 Functional Requirements

To achieve the above-mentioned goal, the ship should comply with the requirements for machinery installations in SOLAS as modified and/or supplemented by the functional requirements of this chapter.

26.2.1 Machinery installations should be capable of delivering the required functionality to ensure availability, including backup functions, sufficient for the autonomous and remotely controlled functions, taking into account connectivity and remote interaction levels to maintain operation in normal and emergency situations.

26.2.2 Machinery installations should be able to support designed [fallback states] and be fault tolerant to connectivity being lost or below an acceptable threshold.

26.2.3 Condition-based monitoring should be provided to assess system reliability.

26.2.4 Measures should be provided to prevent the activation of machinery by remote or autonomous systems when operated or serviced by authorized persons on board.

26.2.5 Measures should be provided to detect machinery malfunctions or failures to maintain safe operation in normal and emergency situations.

26.2.6 Availability and reliability measures should be provided according to the mode of operation to respond to machinery malfunctions and failures.

CHAPTER 27 ELECTRICAL INSTALLATIONS

27.1 Goal

The goal of this chapter is to provide for safe electrical installations.

27.2 Functional requirements

To achieve the above-mentioned goal, the ship should comply with the requirements for electrical installations in SOLAS as modified and/or supplemented by the functional requirements of this chapter.

27.2.1 Power production and distribution should be capable of maintaining the ship in normal operational condition and ensuring that essential systems remain operational in emergency situations for the period(s) specified by SOLAS.

27.2.2 Electrical systems should be able to support designed [fallback states] and be fault tolerant to connectivity being lost or below an acceptable threshold.

27.2.3 Condition-based monitoring should be provided to assess system reliability.

27.2.4 Measures should be provided to prevent electrical systems being energized by remote or autonomous systems when operated or serviced by authorized persons on board.

27.2.5 Measures should be provided to detect electrical system malfunctions or failures to maintain safe operation in normal and emergency situations.

27.2.6 Availability and reliability measures should be provided according to the mode of operation to respond to electrical system malfunctions and failures.

NOTE: The discussion of chapter 28 considering the proposal to delete whole chapter by Norway is postponed to MSC 109.

[CHAPTER 28 EMERGENCY RESPONSE

28.1 Goal

The goal of this chapter is to provide measures for adequate responses in emergency situations.

28.2 Functional requirements

To achieve the above-mentioned goal, the ship should comply with the functional requirements of this chapter.

28.2.1 Emergency response should prioritize the protection of human lives, environment, ship(s) and ROC(s), eliminating or mitigating the impact of the incident and preventing the escalation of the emergency.

28.2.2 Emergency response should be provided to enable the person or system to timely evaluate and decide on the emergency scale and subsequent response level in the event of an emergency.

28.2.3 An adequate communication system with external notification points, including ships in the vicinity, ROC, and ashore, should be maintained [during] [in the event of] an emergency.

28.2.4 Emergency response plan should cover all steps from the detection to the termination of the emergency and the vessel and personnel are in a safe state.

28.2.5 In the event of an emergency, the mode of operation of the ship should be considered in determining the response action.

28.2.6 In the event that the fallback response fails due to deviations from the normal operating status of the ship, emergency response measures should be implemented.]

ANNEX

APPROVAL PROCESS

1 Preliminary design development

The following vessel-specific documentation should be compiled and submitted:

1. Concept of operations (ConOps); as described in chapter 2.1 of the Code.
2. Preliminary design documents; the initial set of preliminary documents should be submitted as deemed necessary to illustrate the main characteristics of the vessel and system arrangements, especially related to autonomous and remotely controlled functions.
3. High level risk assessment report; the objective of the high-level risk assessment is to identify safety critical areas and functions at an early stage and to assist as far as practicable to the drafting of the initial approval basis.
4. Approval basis; should be submitted for approval by the Administration at the end of the preliminary design development. It should be highlighted that within the context of an iterative approval process, it is expected that that this document will be modified significantly throughout the process.
5. Preliminary actions register; the necessary actions for the completion of the process should be drafted, while the early involvement of key stakeholders should be ensured for a clear understanding of roles and responsibilities in the approval process.

2 Preliminary design approval

1. Risk assessment report; A risk assessment, as described in 2.2 of the Code should be performed for all the functions affected by autonomy or remote control. The level of detail of the risk assessment should be proportional to the complexity of the project.
2. Preliminary design documents; As the approval process is an iterative one, the purpose of the preliminary design documents is to further describe and illustrate the key elements of the project that prove the equivalence justification. At the end of this step and possibly after more than one iteration, the design parameters of the systems and system interaction in question should be clear enough to be able to determine appropriate performance criteria that could be verified through testing and other verification methods.

At the end of the step, there should be an alignment between the submitted documents and the risk analysis in terms of assumptions and philosophy, especially regarding mitigation measures. The preliminary design documents could be the following:

- Safety philosophy
- Design philosophy
- Operation and maintenance philosophy
- Emergency response philosophy
- General arrangement

- Systems and Equipment matrix
- ConOps including OE (updated)
- 3. Drawings & information documents (optional); While the iteration process of the project advances, it might be necessary for the approval process to also submit relevant drawings and information documents to clarify certain aspects of the design, especially on issues that are found to be safety critical (i.e. implying very high risk according to the risk analysis). It is expected that such issues that need to be demonstrated at a more detailed level are issues where redundancy, fault tolerance or fail-safe mechanisms need to be further explained.
- 4. Task allocation summary; A task and function allocation summary should be submitted describing the distribution of functions and tasks between human and machine/systems in both normal, abnormal and emergency situations. The task allocation summary should be aligned with the other design documents. There should be particular focus on the expected control actions performed, while especially in the case of human operators it should be made possible to evaluate their expected workload but also their cognitive support.
- 5. Approval basis; it is expected to be updated with each iteration especially when design details and assumptions are decided and documented. At the end of this step the approval basis should be a significantly more detailed document than the one presented in 5.3.1.
- 6. Regulatory gap analysis; A regulatory gap analysis should be submitted to document any deviations from the applicable regulatory framework. At the end of the preliminary approval step, it is not expected that this document will cover all the applicable prescriptive provisions, however it would be beneficial for the detailed design approval to already introduce as much detail as possible. This should be demonstrated through a link between the risk analysis and justification on why design or operational solutions are justified as being equivalent.
- 7. Verification and validation plan; The final step to the preliminary design approval is the detailed definition of how it is intended to perform verification and validation (V&V) of the systems and the MASS as a whole. The objective of the V&V plan should be to describe how functionalities regarding autonomy and remote operations will be verified. A V&V plan should be submitted for approval considering the following:
 - High risk functions and system components as they stem from the risk assessment
 - Boundary conditions and system safety requirements and constraints
 - Cybersecurity related features upon which the protection of the safety critical components is based
 - Incorrect and unexpected inputs and input sequences and timing
 - Reaction of the system-to-system faults and failures
 - Fail-safe modes and fallback states
 - Operational procedures for V&V
 - Ensure that no additional hazards are introduced during V&V

The V&V plan should also include a detailed time frame including intermediate deliverables and reports as deemed necessary. A periodic progress review by the Administration [and any organization performing third-party verification] should be considered. If the complexity of the project does not allow for a complete time frame to be set at this stage, an initial time frame may be accepted for a specific period to be agreed upon.

8. Actions register; The actions register, as described in .3.1.5 should be updated accordingly.

3 Testing and other verification methods

Testing and verification should be conducted according to the defined V&V plan and the relevant reports submitted for information. According to the testing results, the design documents, as well as the approval basis, the actions register and the V&V plan may need to be updated accordingly. In the case that the V&V plan is reflecting an outdated time frame, it should be updated accordingly.

A high degree of transparency with the Administration and any organization conducting third-party verification is strongly recommended to facilitate this approval step.

Different testing and verification methods might be used upon request and should be approved by the Administration. Model tests or simulations are recommended to verify the control system before a full-scale test of the ship is performed.

For tests to be conducted, the relevant procedure(s) should be submitted to the Administration in due time before testing. The test report should be submitted after testing, where a summary or test log should include how the test and its results are linked to the design documents and the V&V plan.

The testing of certain systems or parts of systems may include an element of simulation-based testing. Such testing should not replace full-scale testing.

4 Final approval

This approval step should follow the approach from MSC.1/Circ.1455. It is particularly important to correlate the different components and systems that constitute the MASS in consideration. In addition, this step should be used as a verification that the different steps leading to the final approval are consistent and can be easily verified.

Before the initial survey of the vessel, testing needs to have been completed to demonstrate the requirements in the Approval Basis have been met. This evidence would typically include the final Design Documents and the reports of activities undertaken including a link to the related item of the V&V Plan.

A summary of the equivalence justifications should be submitted. This summary should articulate the approach taken to demonstrating compliance with the Approval Basis and include the Submitter statement that compliance with the Approval Basis has been demonstrated.

The Submitter should propose any survey requirements associated with the system in question. The operational requirements (e.g., training, maintenance) to address the innovative technology aspects should also be included in the In-Service Documentation. Focus should be put on any operational restrictions.

	Preliminary design development	Preliminary design approval	Testing and other verification methods	Final approval	[Survey and Certification]	Operation
Preliminary design documents	X*	X				
Drawings and information documents		X		X		X**
Risk Assessment	X*	X		X		X**
Task allocation summary		X		X		X**
Approval basis and Actions register	X*	X	X	X		X**
Regulatory gap analysis		X				
Verification and validation plan		X				
Testing and verification reports			X			

* Preliminary and high level only

** In case of changes in the approved concept, assumptions and conditions